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27 May 1988
SUPERSEDING
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16 October 1984

MILITARY SPECIFICATION

TRUCK, MAINTENANCE: WITH ROTATING HYDRAULIC DERRICK, AIR TRANSPORTABLE, DIESEL ENGINE DRIVEN, 39,500 POUNDS GVW, 6x4, COMMERCIAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a diesel engine driven, six-wheel, four-rear wheel drive, 39,500 pounds gross vehicle weight (GVW), commercial truck equipped with a maintenance body and a hydraulically operated rotating and telescoping derrick with an insulated pin-on jib or insulated telescopic section for mounting an aerial basket(s). Vehicles procured under this specification are commercial items which shall be warranted by the manufacturer as specified in acquisition documents.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS), and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-UED, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC-2320

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SPECIFICATIONS
FEDERAL

W-B-131	- Battery, Storage: Vehicular, Ignition, Lighting and Starting.
ZZ-T-381	- Tires, Pneumatic, Vehicular (Highway).

MILITARY

MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-H-83282	- Hydraulic Fluid, Fire-Resistant, Synthetic Hydrocarbon Base, Aircraft, NATO Code Number H-537.

STANDARDS
FEDERAL

FED-STD-297	- Rustproofing of Commercial (Nontactical) Vehicles.

STANDARDS
MILITARY

	MIL-STD-1223	- Nontactical Wheeled Vehicles Treatment, Painting, Identification Marking and Data Plate Standards.
*	MIL-STD-1791	- Designing for Internal Aerial Delivery in Fixed Wing Aircraft.
*	MS51118	- Pintle Assembly, Towing - 40,000 Lbs. Capacity, Manual Release.
	MS51317	- Light, Warning, Vehicular, Rotating, D.C.

* 2.1.2 Other Government documents. The following other Government documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

AFSC Design Handbook

DH-1-11	- Air Transportability Handbook.

(Application for copies of the AFSC Design Handbook should be addressed to ASD/ENESS, Wright Patterson AFB, OH 45433.)

DEPARTMENT OF TRANSPORTATION (DoT)
Federal Motor Carrier Safety Regulations.
Federal Motor Vehicle Safety Standards.

(Application for copies of DoT publications should be addressed to the Department of Transportation, Federal Highway Administration, Washington, D.C. 20591.)

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ENVIRONMENTAL PROTECTION AGENCY (EPA)
Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines.
Noise Emission Standards for Transportation Equipment - Medium and Heavy Trucks.

(Application for copies of EPA publications should reference the Code of Federal Regulations, 40 CFR, and the Federal Register and should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)
Occupational Safety and Health Standards.
1926.556 - Aerial Lifts.

(Application for copies of OSHA publications should reference the Code of Federal Regulations, Title 29, and should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

(Copies of specifications, standards and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the non-Government document which is current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)
ANSI A92.2 - American National Standard for Vehicle-Mounted Elevating and Rotating Aerial Devices.

(Application for copies of ANSI publications should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
ASTM Standards.
ASTM Designation: A120 - Pipe, Steel, Black and Hot-Dipped Zinc-Coated (Galvanized) Welded and Seamless, for Ordinary Uses (DoD adopted).
B117 - Standard Method of Salt Spray (Fog) Testing (DoD adopted).

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(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

SAE, INC.
SAE Standards and Recommended Practices

	J318	- Air Brake Gladhand Service (Control) and Emergency (Supply) Line - Couplers, Trucks, Truck-Tractors, and Trailers (DoD adopted).
	J350	- Spark Arrester Test Procedure for Medium Size Engines (DoD adopted).
	J537	- Storage Batteries.
	J551	- Performance Levels and Methods of Measurement of Electromagnetic Radiation from Vehicles and Devices (30-1000 MHz).
	J560	- Seven-Conductor Electrical Connector for Truck-Trailer Jumper Cable.
	J588	- Turn Signal Lamps.
	J589	- Turn Signal Switch.
	J682	- Rear Wheel Splash and Stone Throw Protection (DoD adopted).
	J688	- Truck Ability Prediction Procedure (Dod adopted).
	J704	- Openings for Six- and Eight-Bolt Truck Transmission Mounted Power Take-Offs (DoD adopted).
	J706	- Rating of Winches.
	J844	- Nonmetallic Air Brake System Tubing (DoD adopted).
	J1067	- Seven-Conductor Jacketed Cable for Truck and Trailer Connectors.
*	J1113	- Electromagnetic Susceptibility Procedures for Vehicle Components (Except Aircraft).
	J1292	- Automobile, Truck, Truck-Tractor, Trailer and Motor Coach Wiring.
	J1349	- Engine Power Test Code - Spark Ignition and Diesel.

(Application for copies of SAE publications should be addressed to SAE, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

THE TIRE AND RIM ASSOCIATION, INC.
Yearbook.

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(Application for copies of Tire and Rim Association publications should be addressed to the Tire and Rim Association, Inc., 3200 W. Market Street, Akron, OH 44313.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

* 3.1 Standard vehicle and accessories. Except as specified in 3.1.1 through 3.1.1.14, the vehicle, components, assemblies and accessories to be delivered under the contract shall be standard or optional items, which meet or exceed the requirements of this specification. All chassis items shall be as represented and rated in the chassis manufacturer's technical data and special bodies or mounted equipment shall be as represented and rated in the body and equipment manufacturer's technical data. Technical data shall be limited to specifications and technical material, identical to that furnished to the authorized company representatives for selection of vehicle models and components, and shall be available to the engineering offices of the procuring activity prior to delivery of the items. The chassis model furnished shall be not older than the chassis manufacturer's current model on the date of invitation for bids. When specified (see 6.2), a first article and a first article test shall be provided.

3.1.1 Special requirements. In addition to the standard vehicle and components specified in 3.1, the vehicle shall be furnished with special equipment as specified herein.

* 3.1.1.1 Painting and marking. Treatment, painting, marking and data plates shall be in accordance with MIL-STD-1223. The exterior color and the identification markings shall be in accordance with the requirements of MIL-STD-1223 for the military service identified by the procuring activity (see 6.2). Extruded, pultruded, molded or spiral wound derrick components shall not be painted. When specified (see 6.2), concealed markings shall be furnished.

3.1.1.1.1 Caution plates. Caution plates shall be furnished and mounted in conspicuous locations near the controls and in the driver's compartment. Caution plates shall conform to paragraph 4.10 of ANSI A92.2 and composition A (class 1 or 2) or composition C of MIL-P-514. At least the following information shall appear on the plates:

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WARNING

(a) DERRICK MUST BE IN STOWED POSITION WHILE VEHICLE IS BEING MOVED.
(b) RATED BASKET LOAD, HYDRAULIC OUTRIGGERS DOWN, NOT IN EXCESS OF _____ POUNDS.
(c) INSPECT BASKET LEVELING SYSTEM.
(d) MAXIMUM VOLTAGE DERRICK AND BASKET WILL WITHSTAND _____ VOLTS AC.
(e) OVERALL TRAVEL HEIGHT WHEN DERRICK IS IN STOWED POSITION _____.
(f) VEHICLE SHALL NOT BE OPERATED IN WINDS IN EXCESS OF 30 MPH.
(g) BASKET LINER MUST BE INSTALLED WHEN OPERATING ON OR NEAR ELECTRICAL LINES.
(h) DO NOT PERMIT DIRECT CONTACT OF BASKET TO GROUND WHEN OPERATING ON OR NEAR ELECTRICAL LINES.
(i) WITH OUTRIGGERS PLACED FIRMLY ON THE GROUND, MAXIMUM SIDE SLOPES, DEGREES _____. MAXIMUM ASCENDING SLOPES, DEGREES _____. MAXIMUM DESCENDING SLOPES, DEGREES _____.
(j) WHEN USING PLASTIC NONCONDUCTIVE SECTION AND BASKET(S), REMOVE THE METAL WINCH CABLE AND OTHER CONDUCTIVE ACCESSORIES.
(k) DO NOT DRILL HOLES IN BOOM OR BASKET TO DRAIN WATER, DUMP OUT WATER.

3.1.1.1.2 Instruction plates. Plates showing schematics of all valves, valve controls, equipment, lubrication, and complete operating instructions for the vehicle aerial equipment shall be fastened on the inside of one of the curbside vertical cabinet doors. Each control lever shall be marked with a plate depicting the function of the control. Individual marking shall be adjacent to each control. Plates and markings shall conform to composition A (class 1 or 2) or composition C of MIL-P-514.

3.1.1.1.3 Warning signs. Warning signs or labels as recommended by the Manufacturers of Aerial Devices and Digger Derricks Council shall be furnished and attached.

3.1.1.2 Rustproofing. When specified (see 6.2), the vehicle shall be rustproofed in accordance with FED-STD-297. When specified (see 6.2), tropical rustproofing in accordance with FED-STD-297 shall be furnished.

3.1.1.3 Drain plugs. Drain plugs installed in the manual transmission and the rear axles for draining lubricants shall be of the permanent magnet type.

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3.1.1.4 Wood treatment. Wood shall be treated in accordance with MIL-STD-1223.

3.1.1.5 Towing devices. Not less than two hooks, loops or pins hooks, mounted to the chassis frame or to rigid members which are attached to the frame, for towing the vehicle, shall be furnished. Towing devices shall be furnished at the rear of the vehicle. Towing devices shall be positioned so that the towing cable(s) will not deform other vehicular components. Towing devices may be utilized for the safety chain attachment devices specified in 3.1.1.6.

★ 3.1.1.6 Pintle assembly. A rotating pintle assembly in accordance with MS51118 shall be furnished and installed on the rear of the vehicle. The pintle assembly shall be installed on the vehicle chassis frame with reinforcements to transfer a vertical tongue load of not less than 4,000 pounds and a horizontal drawbar load of not less than 40,000 pounds directly to the chassis rails. The rearmost portion of the pintle shall be forward, but not more than 4 inches forward, of the rearmost part of the vehicle. Two trailer safety chain attachment devices, one adjacent to each side of the pintle, shall be provided. Each attachment device shall provide an ultimate strength of not less than 40,000 pounds. The safety chain attachment devices shall be capable of accepting a standard grab hook (4-9/16 inches wide, 1-3/16 inches thick and 25/32-inch throat width) for a 5/8-inch chain. Safety chain attachments may be utilized for the rear towing devices (see 3.1.1.5).

3.1.1.7 Lighting socket. A trailer lighting socket and cover assembly, conforming to SAE J560, shall be furnished. The connector socket shall be mounted at a readily accessible location on the rear of the vehicle, near the pintle.

3.1.1.8 Trailer lighting cable. A coiled trailer lighting cable, conforming to SAE J1067, shall be furnished. The lighting cable shall be not less than 10 feet long when fully extended. Both ends of the cable shall be equipped with a round plug conforming to SAE J560. The plugs shall be equipped with a grip for withdrawing from the connector sockets. The cable shall be packaged and stowed in the vehicle tool compartment.

3.1.1.9 Electric trailer brake controller. The vehicle shall be provided with an electric brake controller. The controller shall be installed in the truck cab and wired through the SAE J560 12-volt socket. The controller shall operate on the 12-volt electrical system of the vehicle and shall include load control, hand or foot lever, and controller for synchronizing the electric brakes and brake lights of the trailer with the truck brakes.

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3.1.1.10 Spark arrester. As specified herein (see 3.4.4), the vehicle shall be furnished with an exhaust system spark arrester.

* 3.1.1.11 Rear wheel splash and stone throw protection. Rear wheels shall be provided with splash and stone throw protection in accordance with SAE J682. A metal strip not less than 1/8 inch thick and not less than one inch wide, extending the entire width of the mud flap, shall be installed to prevent the bolt heads or bolt nuts from damaging the mud flap.

* 3.1.1.12 Air transportability. The vehicle, complete with all components and accessories and assembled for operational use, shall meet the air transportability requirements of MIL-STD-1791 and AFSC Design Handbook DH-1-11 for transport in C-130 and C-141 aircraft, except only the 36-inch auger need be tied down and shipped with the vehicle during air transport. (Stowage provisions shall be provided for the other augers during over-the-highway use of the vehicle as specified herein.)

* 3.1.1.12.1 Tiedown devices. The vehicle and equipment shall be provided with tiedown devices to permit tying down of the vehicle to the floor or deck of the transportation medium in such a manner as to prevent shifting or movement in any direction. Each device shall be designed to accept its proportionate share of the loads imposed. The devices shall be designed to withstand thrusts specified in MIL-STD-1791 of the following magnitude (G equals weight of vehicle):

(a) Forward thrust 3G
(b) Rearward thrust 1-1/2G
(c) Upward thrust 2G
(d) Downward thrust 4-1/2G
(e) Side thrust 1-1/2G.

Tiedown devices shall be located so as to permit easy attachment of cables or turnbuckles. Tiedown devices shall be indicated on an information plate conforming to type III, composition A or C of MIL-P-514. Marking shall clearly indicate that the devices are intended for tying down the vehicle on the carrier when shipped.

3.1.1.13 Radiator shutters. When specified (see 6.2), radiator shutters shall be furnished. The radiator shutters shall be thermostatically controlled, air actuated, and of the multivane type. The radiator shutters shall maintain the correct coolant temperature. Means shall be provided for manual operation. The shutters shall remain in the open position should the thermostatic controls fail.

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* 3.1.1.14 Vehicle keys. When specified (see 6.2), keys common to all trucks shall be provided for the ignition and all doors.

3.2 General design.

3.2.1 Federal Motor Vehicle Safety Standards. The vehicle and furnished accessories shall comply with all Federal Motor Vehicle Safety Standards in effect on the date of manufacture.

3.2.2 Air pollution control. The vehicle shall comply with EPA Regulations governing control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines in effect on the date of manufacture. In addition, vehicles destined for California shall comply with State of California regulations governing air pollution control in effect on the date of manufacture.

3.2.3 Sound level. The interior sound level shall conform to Federal Motor Carrier Safety Regulation 393.94. The vehicle exterior sound level shall conform to EPA Noise Emission Standards for Transportation Equipment, Medium and Heavy Trucks.

3.2.4 Curb weight. The curb weight shall include the weight of chassis and cab, with all attachments, accessories and equipment; a 36-inch auger stowed on the vehicle for air transport and the other augers not on the vehicle; the body; and full complement of fuel, lubricants and coolant. The curb weight shall not exceed the load limit in C-130 and C-141 aircraft as specified in 3.1.1.12. The curb weight on the front axle shall not exceed 13,000 pounds. The curb weight on the rear tandem shall not exceed 26,000 pounds.

3.2.5 Gross vehicle weight. Gross vehicle weight (GVW) shall consist of the curb weight, operator weight (computed at 175 pounds), and a payload evenly distributed in the vehicle body load area to provide not less than the specified GVW. Minimum GVW shall be not less than 39,500 pounds.

3.2.5.1 Weight distribution. The axle capacity and the capacity of the suspension and tires per axle shall be not less than the load imposed with the vehicle loaded as specified in 3.2.5 and in no case less than 12,000 pounds for the front and 30,000 pounds for the rear. In addition, the capacity of the front axle and suspension system and the capacity of the rear axle and suspension system shall be not less than 1-1/4 times the loads imposed by the curb weight of the vehicle with the 36-inch auger tied down on the vehicle for air transport.

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3.2.6 Ratings. The vehicle and component ratings shall be manufacturer's published ratings. Component and vehicular ratings shall not be raised to meet the requirements of this specification. When ratings are not published, verification of ratings shall be available to the engineering office of the procuring activity.

* 3.2.7 Safety. The vehicle and rotating derrick shall conform to ANSI A92.2 and to all OSHA requirements, including 1926.556.

3.2.8 Overall width. The overall width of the vehicle, exclusive of mirrors, lights, reflectors and tires shall be not more than 96 inches. The width over the tires shall be not more than 100 inches.

* 3.2.9 Accessibility. The design of the vehicle and optional equipment shall permit access for routine servicing and shall permit access for replacement and adjustment of component parts and accessories with minimal disturbance of other components and systems.

3.3 Performance.

3.3.1 Speeds and gradeability. High and low speed requirements shall be met at a gross combined weight (GCW) of 60,000 pounds with the vehicle loaded to the specified GVW.

3.3.1.1 High speed gradeability. The vehicle (at 60,000 GCW) shall ascend a continuous grade of not less than 0.25 percent at 50 miles per hour (mph). Gradeability requirements shall be met with the main transmission in direct drive, and when multispeed axle or auxiliary transmission is furnished, with the axle or auxiliary transmission in high speed ratio. Gradeability shall be verified with calculations in accordance with SAE J688 (see 6.3).

3.3.1.2 Low speed. Low speed shall be calculated with engine operating at not less than 35 percent of the recommended governed speed, and shall provide a vehicle speed of not more than 3.5 mph.

3.3.1.3 Maximum geared speed. The maximum geared speed at engine governed speed shall be not less than 58 mph. Conformance to geared speed specified shall be determined by calculating in accordance with the following formula:

$$\text{Maximum geared speed (mph)} = \frac{\text{Governed speed (rpm)}}{\text{Total gear reduction x tire factor (see 6.3)}}$$

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3.3.2 Service brakes. The service brakes shall control and hold the vehicle, when loaded to its specified GVW, on a 30 percent grade. The service brakes shall stop the vehicle, loaded to specified GVW, within the stopping distance requirements of Federal Motor Carrier Safety Regulation 393.52.

3.3.3 Height and reach. The derrick shall provide for hydraulically powered movement from below horizontal to not less than 80 degrees above horizontal and return, continuous rotation about a vertical axis in both directions, and telescoping extension and retraction of all metal sections. For each telescoping metal extension position listed in table I at not more than the corresponding angle specified in table I, the derrick height from the ground to the centerline of the upper head sheave shall be not less than that specified in table I. The basket floor-to-ground height shall be not less than specified in table I.

TABLE I. Height, reach and loads.

Telescoping section position	Degrees from horizontal	Height from ground (feet)	Load (pounds)
Without fiberglass jib or with the insulated telescopic section retracted:			
1. All telescoping metal extensions retracted	80	31 max.	15,000 min.
2. All telescoping metal extensions extended	80	60 min.	7,000 min.
3. All telescoping metal extensions extended	60	54 min.	1,000 min.
4. All but the outermost telescoping metal extension(s) extended	60	40 min.	2,000 min. 1/
5. All but the outermost telescoping metal extension(s) extended	45	manufacturer's standard	1,000 min. 1/
6. All but the outermost telescoping metal extension(s) extended and boom horizontal or with the insulated telescopic section extended	0	7-1/2 min.	1,000 min. 1/, 2/

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TABLE I. Height, reach and loads. - Continued

Telescoping section position	Degrees from horizontal	Height from ground (feet)	Load (pounds)
With fiberglass jib in place or with the insulated telescopic extension extended:			
1. Without baskets, with all telescoping metal extensions extended	80	80	1,500 min.
2. Without baskets, with all telescoping metal extensions extended	60	70 min.	800 min.
3. Without baskets, with all but the outermost telescoping metal extensions extended and boom horizontal	0	7-1/2 min.	600 min. 1/
4. With baskets, with all telescoping metal extensions extended; basket floor-to-ground height (one basket)	80	76 min.	300 min.
	60	65 min.	300 min.
(two baskets)	80	76 min.	600 min.
	60	65 min.	600 min.
5. With baskets, with all but the outermost telescoping metal extensions extended; basket floor-to-ground height (one basket)	60	45 min.	300 1/
(two baskets)	60	45 min.	600 1/

1/ The derrick shall be provided with limiting devices set to restrict down-movement of the fully extended telescopic boom, with jib and basket in place or with the insulated fiberglass section extended and basket in place; and also without the jib and basket or with the insulated fiberglass section retracted; into unstable, overloaded or inadequate hydraulic power positions. Retraction of the outermost telescopic metal extension(s) then further lowering shall, however, be possible. The position at which down-movement automatically stops and retraction of the outer section must take place shall be not more than 60 degrees above horizontal.

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2/ When a four-section metal telescoping boom is furnished, the second outermost telescoping metal extension may be retracted to achieve the specified load, provided the baskets can still reach the ground, and provided limiting devices are furnished which are set to restrict down-movement until the two outermost metal extensions are retracted. The position at which the second outermost telescoping metal extension must be retracted shall be not more than 45 degrees above horizontal.

3.3.4 Vertical loads. For each position listed in table I, with the outriggers extended, the derrick shall be capable of suspending, on the winch line over the upper head sheave, the corresponding load at not more than the specified angle and at all greater angles attainable. The derrick shall be capable of elevating from and lowering back to the angle specified with the load but shall be incapable of elevating from a horizontal position with loads that would result in permanent deformation. The derrick shall be capable of retracting (from extended positions) and returning back to extended position with the load. All operations of rotating, elevating or lowering, and extending or retracting, shall be possible singly and in all combinations, all while both loaded and unloaded. Operations as specified above shall be possible at all rotational positions of the derrick without any vehicle wheel leaving the ground and without the permanent deformation of any part. The loads specified in table I shall be in addition to the loads imposed by the digger with a 24-inch auger in place and by the pole grabber.

3.3.5 Rotation. The derrick assembly shall be capable of continuous rotation in either direction about a vertical axis. The turret assembly shall rotate the loads specified in 3.3.4 through an arc of 360 degrees at a speed of not less than 1/2 revolution per minute (rpm) and not more than 1-1/2 rpm. The turret assembly shall be capable of smoothly rotating the derrick with the vehicle up and down a 15 percent grade at all derrick positions and loads listed in table I without any vehicle wheel leaving the ground and without permanent deformation of any part.

3.3.6 Horizontal load. The turret and derrick shall be capable of rotating against a horizontal load, at right angles to the derrick, of sufficient force to produce a torque of not less than 42,000 foot-pounds without opening the turret rotation hydraulic relief valves and without permanent deformation of any part. The turret shall be provided with hydraulic relief valves preventing rotation against a horizontal load, at right angles to the derrick, that would result in permanent deformation of any part.

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3.3.7 Minimum height. With the outermost metal section of the boom retracted, the boom shall be capable of lowering the loaded basket to the ground over the rear of the vehicle.

3.3.8 Position stability. The turret and derrick assembly shall be capable of remaining in a fixed position and of controlled lowering when the hydraulic motor is de-energized. With the vehicle longitudinal on a 15 percent grade without being leveled and the derrick transverse, the turret and derrick shall hold in all positions and load combinations specified in 3.3.4 without more than 1 inch per hour horizontal creep and without more than 2 inches per hour vertical settling, both measured at the centerline of the upper head sheave.

3.3.9 Upsetting stability. With 1-1/2 times the load specified in 3.6.3.3 in the baskets, the derrick shall perform as specified in 3.3.3 through 3.3.7 through all positions the derrick is capable of attaining without causing the vehicle to become unstable. The vehicle shall be deemed unstable with 1-1/2 times the specified loads when any one of the vehicle wheels or outriggers lifts off of the ground. The vehicle shall also be deemed unstable if with twice the specified load (see 3.6.3.3), the inner tire of either dual rear wheel lifts off of the ground. The stability test shall not cause any permanent deformation beyond the elastic limit of any component of the vehicle.

* 3.3.10 Digger location. The digger shall store on the main boom (first stage) and transfer to the next section of the boom (second stage). At 0 degrees, the horizontal radius, from the centerline of the turret to the centerline of the digger, shall be variable between at least 21 to 31 feet. The telescopic boom capacity, in these positions, with the telescopic boom horizontal and all accessories in place, shall be not less than 800 pounds.

3.4 Chassis components.

3.4.1 Engine. The engine furnished shall be the chassis manufacturer's standard or optional engine for the commercial model vehicle which meets or exceeds the requirements of this specification.

3.4.1.1 Diesel engine. The vehicle shall be equipped with a liquid cooled, compression ignition, two-stroke or four-stroke cycle diesel engine, with not less than six cylinders. Engine net horsepower figures used in performance prediction calculations shall be determined in accordance with SAE J1349.

3.4.1.2 Oil filter. A full flow type oil filter shall be furnished.

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3.4.1.3 Governor. An engine governor shall be furnished and set and sealed to limit the engine to the engine manufacturer's maximum recommended operating speed.

* 3.4.1.4 Cooling system. The cooling system shall maintain the engine coolant at a temperature below the boiling point with the vehicle loaded to rated GVW and operated at an altitude of 10,000 feet above sea level or in an ambient air temperature of not less than 125 degrees Fahrenheit (°F). The coolant system shall include a surge tank or a coolant recovery reservoir of not less than two-quart capacity. A high coolant temperature or a low coolant level alarm buzzer and a red indicator warning light shall be provided on the instrument panel.

3.4.1.5 Coolant temperature control. Thermostatic control of the engine coolant temperature shall be provided. The control shall include partial thermostatic control of coolant flow through the radiator and thermostatically controlled radiator shutters, or complete thermostatic control of all coolant flow through the radiator.

* 3.4.1.6 Fan clutch. A fan clutch shall be provided. The fan clutch shall reduce the fan speed automatically when the fan is not required for engine cooling.

3.4.1.7 Silicone rubber hoses. When specified (see 6.2), silicone rubber radiator and heater hoses shall be furnished.

3.4.1.8 Power plant heaters and fuel warmer. When specified (see 6.2), a coolant heater, an engine oil heater, and a fuel warmer shall be provided. Unless otherwise specified (see 6.2), a battery heater shall be provided when power plant heaters are specified. The heaters shall operate on 120-volt alternating current (ac) and shall be wired through a junction block to a single three-pronged (male), weatherproof slave receptacle for receiving external power and grounding the vehicle. A three-wire connecting cable, 25 feet long and of adequate line capacity to supply power for all heater units simultaneously, shall be furnished. The connecting cable shall include a matching female connector at the vehicle end and a standard, three-pronged (two power plus one ground) male connector at the other end. Electrical apparatus shall conform to Federal Motor Carrier Safety Regulation 393.77(c)(7). Electrical insulation of connecting cable shall withstand normal operating stresses in low ambient air temperatures (down to minus 60°F) without cracking or loss of dielectric capacity. All heater lead wires shall be installed without interfering with vehicle component operation, and without loose excess wire. Provisions for stowage of the cable shall be provided in the vehicle cab. Heaters shall be furnished as follows:

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(a) Coolant heater, 1500-watt minimum rating, shall be installed in the engine block or in the lower coolant inlet hose. Engine thermostat with an operating range of 170° to 195°F shall be installed

(b) Immersion type engine oil heater, 300-watt minimum rating, with 170° to 195°F thermostat, shall be installed in oil pan through any convenient opening

(c) Battery heater shall have a capacity adequate to maintain the battery electrolyte at a temperature of not less than 10°F during vehicle exposure in ambient air temperatures as low as minus 60°F, and shall embody a thermostat to limit the temperature of the electrolyte to not more than 80°F

(d) A fuel warmer or preheater shall be provided to prevent clogging of fuel filters due to wax crystallization in the fuel. The fuel warmer shall use engine coolant to transfer sufficient heat to the diesel fuel to heat it from an inlet temperature of minus 40°F to an outlet temperature of plus 9°F, with a flow rate not less than the maximum fuel demand of the engine fuel system. A coolant shutoff valve shall be provided for the coolant inlet side of the fuel warmer unit.

3.4.2 Electrical systems. The electrical systems shall be in accordance with Federal Motor Carrier Safety Regulations 393.27 through 393.31 and 393.33.

* 3.4.2.1 Starting system. A 12- or 24-volt direct current (dc) starting system with a 12-volt lighting system, and with not less than a 80-ampere alternator, shall be furnished. Engine starting equipment shall include an ether starting system or a glow plug. If an ether system is furnished in lieu of a glow plug, it shall be of the measured shot type. The measured shot type ether system shall be key operated or manually operated from the driver's compartment and shall be inoperative with the engine warm. Complete provisions for a replaceable ether reservoir of not less than 12 fluid ounces shall be furnished. A reservoir need not be furnished.

* 3.4.2.2 Lighting. All vehicle lights, reflectors and wiring shall conform to Federal Motor Carrier Safety Regulations 393.12, 393.19, 393.20 and 393.22 through 393.26(d). When available as the manufacturer's standard or optional feature, the brake lights shall override the four-way emergency flasher. Lights and reflectors shall not be mounted on vertical surface of rub rails or mounted on the vehicle bumpers (unless recessed and fully protected).

3.4.2.3 Turn signals. Turn signal lamps shall conform to SAE J588. Operating units shall conform to SAE J589 class A and shall be mounted on the steering column. The vehicle shall be provided with

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double-faced front signal units and with single-faced rear signal units installed in accordance with SAE J588, except that turn signals shall not be mounted on the engine compartment hood. Turn signal operating units shall have a visible and audible type flash indicator.

* 3.4.2.4 Batteries. Each battery shall be of 12-volt potential. The total reserve capacity rating shall be not less than 320 minutes. The total cold cranking ampere rating at 0°F shall be not less than 1,200 amperes for vehicles with less than 220 gross horsepower engines, and 1,740 amperes for vehicles with engines of 220 gross horsepower and over. Reserve capacity and cold cranking ratings shall be measured in accordance with SAE J537. The batteries shall be of the maintenance-free type having the maintenance-free characteristics listed in W-B-131.

3.4.2.5 Radio interference suppression. The vehicle shall be suppressed to limit external electromagnetic radiation in accordance with SAE J551. Any body equipment emitting radiation shall be suppressed to the same level as the vehicle chassis.

3.4.3 Fuel system. The fuel system shall conform to Federal Motor Carrier Safety Regulations 393.65 and 393.67.

* 3.4.3.1 Air cleaner. An engine air cleaner shall be furnished.

* 3.4.3.2 Fuel tank(s). Fuel tank(s) shall be of not less than 100 gallons total capacity (85 gallons for vehicles destined for California). When more than one tank is furnished, means shall be provided to assure an equal fuel level in each tank.

3.4.4 Exhaust system. The exhaust system shall conform to Federal Motor Carrier Safety Regulation 393.83. A spark arrester shall be furnished, except on vehicles with turbocharged engines. The spark arrester shall have an 80 percent arresting efficiency when rated in accordance with SAE J350. The exhaust outlet shall be located to direct the exhaust gases away from the rotating derrick control console in all positions of rotation. The exhaust outlets shall be to the sides of the vehicle, immediately in front of the rear wheels. The outlets shall direct exhaust fumes in a horizontal direction. A vertical exhaust system shall not be furnished.

3.4.4.1 Exhaust system safety shield. The exhaust system or hydraulic system shall be shielded so that, in the event of any hydraulic line rupture or hydraulic fluid leak, the hydraulic fluid will not come into contact with any part of the vehicle exhaust system.

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3.4.5 Transmission. A manually shifted transmission shall be provided on the vehicle. Input torque capacity of the transmission shall be at least equal to the maximum torque delivered by the engine. The vehicle shall be provided with not less than seven forward speeds. Multispeed tandem axle or multirange or auxiliary transmission shall be provided. Gear ratios in the transmission and axles shall be matched to provide a progressive shifting pattern throughout the complete range, and shall provide the required vehicle performance (see 3.3.1 through 3.3.1.3). The transmission shall be provided with two power takeoff openings conforming to SAE J704. The transmission shall provide for maximum ease of shifting in all speeds.

3.4.5.1 Clutch. The clutch shall be the largest capacity clutch offered for the vehicle and engine furnished, with the clutch torque capacity exceeding the maximum delivered engine torque.

3.4.5.2 Power takeoff. The power takeoff shall be of a rated capacity to operate powered equipment. Controls to operate the power takeoff shall be located in the truck cab accessible to the seated driver. A caution plate or decal reading "DO NOT OPERATE VEHICLE AT HIGHWAY SPEEDS WITH POWER TAKEOFF ENGAGED" shall be installed so as to be readily visible to the seated driver.

3.4.6 Driveline components. The driveline components shall be adequate to transmit the maximum delivered torque of the engine, as developed through the maximum gear train reduction.

3.4.7 Frame. The vehicle shall have a heavy duty main frame or frame reinforcements. The heavy duty frame shall have frame rails of greater section modulus than the manufacturer's standard frame and shall provide structural strength at least equivalent to the reinforced frame specified herein. Reinforcements shall extend at least from the rear of the front suspension, rear hanger bracket to the bogie trunnion mounting bracket. The frame, with reinforcements, shall provide structural integrity to adequately distribute and support the additional loads imposed by the rotating hydraulic derrick and the outrigger supports, with the vehicle loaded to its specified GVW. The frame and reinforcements shall have a combined total resisting bending moment not less than that recommended by the hydraulic derrick manufacturer. Chassis frame rails shall not project beyond the rear of the body.

3.4.8 Suspension. The vehicle shall be equipped with a suspension system with components having a rated capacity at least equal to the load imposed on each member, measured at the ground, with the vehicle loaded to the required GVW. In addition, the rated capacity of the front and rear suspension shall be not less than that specified in

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3.2.5.1. When the spring capacity is rated at the spring pads, unsprung weight shall be deducted. The vehicle shall be equipped with hydraulic, double-acting shock absorbers at the front wheels when shock absorbers are available as standard or optional equipment with the suspension system and axle furnished. Suspension system ratings shall be adequate for the loads imposed during all operating positions of the rotating hydraulic derrick.

* 3.4.9 Axles. Axle ratings shall be at least equal to the load imposed, measured at the ground, with the vehicle loaded to the required GVW. In addition, the axle ratings shall be at least equal to that specified in 3.2.5.1.

3.4.9.1 Rear bogie. The rear bogie on the vehicle shall be of the four-wheel type, complete with axles, springs, torque rods and all other necessary parts. The bogie shall be provided with means permitting differential action between the two axles, and a manually or automatically controlled lockout assuring equal power to each rear axle. Manual lockout control used shall be located in the truck cab. Axle gear ratios shall provide the performance specified in 3.3.1 through 3.3.1.3.

3.4.9.2 Traction control. When specified (see 6.2), the vehicle shall be furnished with a traction control on either rear axle. The traction control shall actuate automatically to ensure that power is transmitted to the wheel having traction when the opposite wheel loses traction. Maximum traction capabilities shall be maintained at all times under each drive wheel for the life of the vehicle.

* 3.4.10 Wheels, rims, tires and tubes. The vehicle shall be equipped with single front and dual rear wheels. Rims and tire ratings shall conform to Tire and Rim Association recommendations for the type and size of tires furnished. Tires, rims, and disc wheel sizes shall be the same for all wheels on the vehicle. When specified (see 6.2), ten-stud disc type wheels shall be furnished.

3.4.10.1 Tires. Tires shall be steel belted radial or, when specified (see 6.2), bias ply. Tires shall be tube or tubeless type with highway tread on the front axle and with nondirectional mud and snow tread on the rear axles. Tires shall be of a rated capacity at least equal to the load imposed on each tire, measured at each wheel at the ground, with the vehicle loaded to specified GVW. In addition, tire rated capacity shall be at least equal to that specified in 3.2.5.1. Tires shall conform to Tire and Rim Association recommendations, or to ZZ-T-381 with a size designation system the same as the Tire and Rim Association.

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3.4.10.2 Inner tubes. For tube type tires, inner tubes shall be of the heavy duty type and shall be of the proper size for the tires furnished. Tire flaps shall be provided for tube type tires in accordance with Tire and Rim Association recommendations.

3.4.10.3 Spare wheel or rim. When specified (see 6.2), a spare wheel or rim shall be mounted on a carrier and mounted behind the vehicle cab or behind the derrick in the body loadspace. Threaded fasteners, when used to secure the spare tire in the carrier, shall be constructed of or plated with corrosion-resistant material. The carrier design shall enable removal or mounting of the spare wheel assembly using only the tools specified in 3.4.16.1. When specified (see 6.2), a spare wheel or rim shall be furnished and shall be stowed on the vehicle for shipment. When a spare tire assembly is specified (see 3.4.10.3.1), a spare wheel or rim shall be furnished.

3.4.10.3.1 Spare tire assembly. When specified (see 6.2), a spare tire assembly shall be furnished. The spare tire assembly shall include an inflated spare tire mounted on the spare wheel or rim. The spare tire shall be of the same size and tread design and of the same load range (ply rating) as the tires furnished on the vehicle.

* 3.4.11 Brakes. The brakes shall conform to Federal Motor Carrier Safety Regulations 393.40 through 393.42(b), 393.43 and 393.45 through 393.52. When available as the manufacturer's standard or as an option, brake linings shall be of nonasbestos material.

3.4.11.1 Service brakes. The vehicle shall be equipped with full-air brakes on all wheels. The braking system, complete with all necessary components, shall include:

(a) Air compressor, unloader-head type, engine driven and engine lubricated, air- or water-cooled and having a capacity of not less than 12 cubic feet per minute (cfm)
(b) Air storage reservoir(s), each tank equipped with a drain, and with safety and check valves between the compressor and the last reservoir tank
(c) Foot control, suspended or treadle type
(d) Air control valves
(e) Air pressure gage, visible to the driver
(f) Low air pressure warning, visible and audible
(g) Service brake stop lamp switch
(h) Automatic moisture ejector
(i) Automatic slack adjusters.

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* 3.4.11.1.1 Air dryer. A replaceable cartridge desiccant type air dryer shall be installed in the air brake system. The dryer shall have the capability of removing not less than 95 percent of the moisture in the air being dried. The dryer shall have a precooler and a filter to screen out oil and solid contaminants. The dryer shall have an automatic self-cleaning cycle and a thermostatically controlled heater to prevent icing of the purge valve.

3.4.11.2 Trailer brake control system. In addition to the components specified in 3.4.11.1 and 3.4.11.1.1, a trailer brake control system shall be furnished. The trailer brake control system shall include:

(a) Identification of emergency and service lines
(b) Coincident control of the trailer brakes with the prime mover foot control
(c) Independent hand control for the trailer brakes
(d) Prime mover protection valve with dash control and automatic breakaway feature
(e) Trailer stoplight control operable with the foot brake and with the hand control for trailer brakes
(f) Two SAE J844 coiled air hoses, not less than 110 inches long when fully extended, with SAE J318 gladhand couplers on both ends of hoses. The hoses shall be packaged and stowed in the vehicle tool compartment for shipment
(g) Air connectors for trailers with SAE J318 gladhand couplers mounted at the rear of the vehicle, located to prevent interference with trailer
(h) Dummy gladhand couplers with chains
(i) Prime mover only parking brake valve to permit prime mover parking brakes to be applied while charging the trailer air brake system.

* 3.4.12 Cab. The chassis manufacturer shall furnish any type of his standard or optional full-width cab. When specified (see 6.2), a cab with a forward tilting hood and fender assembly, including tilting and locking mechanism, shall be furnished. When the vehicle is complete with body and rotating derrick components and accessories, the unladen overall height of the cab and the vehicle shall not exceed 102 inches. The overall height shall be measured with the tires inflated to the Tire and Rim Association recommendations for the tire loadings and tires furnished. The cab doors shall be equipped with locks, operable from inside the cab through mechanical linkage, and with at least the curbside door equipped with an external, key operated lock. Drip rails shall be provided above the cab doors. The cab shall have an upholstered, full-width, adjustable seat and back or an individual,

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adjustable driver's seat and an individual passenger's seat. When specified (see 6.2), the cab shall be equipped with an individual, adjustable, driver's seat and individual passenger's seat. The color of the upholstery and the interior finish shall be compatible with the exterior color (see 3.1.1.1). White upholstery shall not be furnished. Safety grips shall be provided on each side of the cab to assist personnel in entering and leaving the cab. Interior lighting shall be provided. Manufacturer's standard fastenings and seat belts shall be furnished for each occupant.

3.4.13 Steering. Power steering shall be furnished.

3.4.14 Windshield wipers and washers. The vehicle shall be equipped with dual windshield wipers and windshield washers. Windshield wipers shall be multispeed type operated by air or electric motor(s).

3.4.15 Rear end protection. The rear of the vehicle shall be protected in accordance with Federal Motor Carrier Safety Regulation 393.86. A rear bumper shall be provided (see 3.5.1).

3.4.16 Stowage space. A stowage space of sufficient size to accommodate a vehicle jack, hand tools, antiskid chains and emergency reflective triangles shall be furnished for retaining equipment during vehicle operation. The stowage space shall be weatherproof and shall provide for locking with a padlock.

3.4.16.1 Tools. When specified (see 6.2), the vehicle shall be furnished with tools required for exchanging a mounted tire assembly with the spare assembly (see 3.4.10.3), and shall include at least a hydraulic jack, jack handle and wheelnut wrench. The jack shall be of such closed height as to permit its location under any axle, or other satisfactory lift point, at any wheel with the tire flat. The jack, without blocking, shall be capable of raising any wheel of the loaded vehicle to a height adequate to permit removal and replacement of the wheel and tire assembly.

3.4.17 Heater and defroster. A heavy duty hot water heater shall be provided. The heater shall have fresh air intakes. Discharge outlets shall be provided to direct heated air to the cab floor and to defroster louvers. The heater shall be complete with a blower and mounted controls convenient to the driver.

3.4.18 Controls and operating mechanisms. All controls and operating mechanisms shall be located for left hand drive. Controls shall be complete and conveniently operable by the driver. Lever controls shall be designed and located to permit easy entrance and exit

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of the operator to and from the driver's compartment. Instruments and controls shall be identified as to their function and installed in a manner to facilitate removal and servicing. The instruments shall be panel mounted.

3.4.19 Accessories and equipment. Chassis equipment shall be complete with all accessories furnished as standard equipment by the manufacturer. The following minimum equipment shall be furnished:

(a) Key-operated ignition switch
(b) Ammeter, charging indicator or voltmeter
(c) Fuel gage
(d) Oil pressure gage or indicator
(e) Engine temperature gage or indicator
(f) Speedometer with recording odometer
(g) Tachometer
(h) Ash receptacle
(i) Dual sunvisors
(j) Armrest on driver's side
(k) Driver's compartment ventilator other than window
(l) Instrument lighting with dimmer control
(m) Shaded dashlight or domelight for map reading.

* 3.4.20 Rearview mirrors. Outside rearview mirrors shall be mounted on each side of the cab. The mirrors shall be of the combination type having flat and convex areas enclosed in a common housing. The flat portion shall have not less than 72 square inches of reflective area. The convex portion shall have not less than 24 square inches of reflective area. The mirrors shall have not less than two supporting arms. Mirrors shall be capable of being swung inboard to a stowed position.

3.4.21 Horn. The manufacturer's standard electric horn shall be furnished. When specified (see 6.2), in addition, an air operated horn shall be furnished.

3.4.22 Engine hour meter. An engine hour meter having a totalizing mechanism of not less than 9,999 hours shall be furnished for the chassis engine to register accurately the number of hours of operating time. The meter shall be of rugged construction to insure continuous trouble-free performance under severe operating conditions. The engine hour meter shall be mounted on the cab instrument panel or in the engine compartment in a readable location.

3.4.23 Backup alarm. An audible, pulsating, signaling device (mechanical or electrical) shall be provided to caution personnel when the vehicle is in reverse gear operation.

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* 3.4.24 Front mounted winch. The vehicle shall be furnished with a hydraulic winch assembly, mounted on the front of the vehicle. The winch shall conform to SAE J706. The winch shall be of the single drum type for hoisting and negotiating purposes and shall have one forward and one reverse speed. Dual winch controls shall be furnished, one set inside the cab and one set at the front bumper. Cab winch controls shall be located to provide no interference with the entrance or exit of the driver. The winch shall have a rated single line pull capacity of not less than 20,000 pounds on the bare drum. The winch shall be equipped with a hydraulic relief valve set to restrict the line pull capacity to 22,000 pounds. The winch line speed on the bare drum shall be between 15 and 30 feet per minute at an engine speed equal to 35 percent of engine governed speed. The winch shall be wound with not less than 185 feet of 5/8-inch diameter, preformed 6x37, improved plow steel, independent wire rope core (IWRC) wire rope and shall be equipped with an end chain and hook. An integral, adjustable, automatic safety brake shall be provided. The winch shall be equipped with a 4-way roller guide assembly. The winch shall be equipped with a shaft extension on the curbside with a capstan drive extension (AT&T type).

3.4.24.1 Winch drum guard. A winch drum guard shall be furnished. The guard shall confine the cable to the area between the drum flanges. The guard shall consist of not less than 1/4-inch vertical side plates, conforming to the outside radius of the drum flanges. Six bars, 3/8 inch by 1-1/4 inches, shall be welded to the vertical side plates. Three bars shall be located on the top and spaced equally on the top radius, and three bars shall be located on the bottom and spaced equally on the bottom radius. The vertical distance between the vertical side plates and the drum flanges shall be not more than half the specified cable diameter.

3.4.24.2 Front bumper. The bumper shall be mounted forward of the winch. The open area on either side of the winch shall be covered with a combination step plate and gravel guard. The bumper shall be either a channel or a pipe type. When a pipe type front bumper is furnished, the nominal diameter shall be not less than 3 inches, and the wall thickness shall be not less than that specified in Schedule 40 of ASTM A120. The pipe type front bumper shall have half-round ball ends.

* 3.4.24.3 Combination step plate and gravel guard. A combination step plate and gravel guard shall be furnished. The step plate shall be fabricated of not less than 14 gage (0.0747 inch) steel tread plate exclusive of projections. The step plate shall be secured to the front bumper. The step plate shall be not less than 1/4 inch and not more than 3/8 inch from the chassis sheet metal or, if the space is greater, a formed rubber intrusion may be installed to seal off the space between

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the step plate and the chassis sheet metal. The step plate shall be capable of supporting 300 pounds per square foot. The step plate shall not deflect more than 1/8 inch under the loads imposed.

3.5 Body. A line maintenance and construction truck body for heavy duty work shall be furnished. The body shall have the dimensions specified in table II. The cab to centerline of the bogie shall be as specified in table II. The body assembly shall include: cabinets on each side of a center loadspace with cabinet doors opening from the outside; hydraulically operated turret and telescopic derrick; rotating hydraulic digger, outriggers, pole grabber, tamper and puller; nonconductive manually placed jib or nonconductive telescoping section; fiberglass basket(s); and other equipment as specified herein. The turret of the derrick shall be located in the loadspace. Cabinet space may be eliminated to accommodate radial outriggers. The cabinet area that is cut to provide clearance for A-frame telescopic type outriggers or rear mounted winch shall be covered with sheet metal to present a completely enclosed and weatherproof cabinet interior. Safety tread steel or aluminum plate shall be installed in the body loadspace, on the top of the cabinets, on the access steps and on the bumper and platform extension. The floor of the loadspace shall be capable of supporting a live load of 500 pounds on any square foot of area without permanent deformation of the floor.

TABLE II. Dimensions and loading requirements.

	Feature	Inches
	Cab-trunnion dimension (minimum)	160
*	Overall height with derrick in stowed position (maximum)	102
	Overall width of body (maximum)	96
*	Overall height of vehicle (maximum)	102
	Overall length of vehicle (maximum) with front winch	470
	Loadspace:	
	Width (exclusive of wheelwells) (minimum)	56
	Height (minimum)	19
	Cabinet space (from rear of cab):	
	Length (including rear platform extension) (minimum)	190
	Height at front (minimum)	40
	Height at rear (minimum)	40
	Width (minimum)	16
	Rear platform (minimum)	30

3.5.1 Bumper and platform extension. A combination bumper and platform extension shall be provided. Safety tread steel or aluminum plate shall be provided on the top of the platform extension.

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3.5.1.1 Rear access steps. Rear access steps shall be furnished below the combination bumper and platform extension. A grab handle(s) shall be installed to assist personnel in entering the body loadspace.

3.5.1.2 Front access step. An access step and grab handle shall be furnished on the curbside of the body. The access step shall be at the forward end of the body.

3.5.2 Body mounting. The derrick shall be attached to a full length subframe extending from immediately behind the truck cab to the end of the chassis frame rails. The full length subframe shall be secured by brackets bolted to the chassis frame rails. The outriggers shall be attached to the derrick or to the full length subframe. No part of the body (cabinets, floor, derrick, turret, outriggers, or subframe brackets) shall be welded to the chassis frame. The mounting of the cabinets and the loadspace floor shall conform to 3.5.2.1 through 3.5.2.1.3 or to 3.5.2.2, at the manufacturer's option.

3.5.2.1 Flexible mounting for cabinet-floor assembly. The cabinets and the loadspace floor shall be an integral assembly, secured independently of the derrick and outriggers to the chassis frame or derrick subframe with U-bolts, twin studs or brackets and shall include a wood breaker strip. All but the rearmost cabinet-floor assembly mountings shall be spring loaded to provide for movement of the cabinet floor assembly independently of the subframe, chassis frame, derrick, turret and outriggers.

3.5.2.1.1 U-bolts or twin studs. When U-bolts or twin studs are used for mounting the cabinet floor assembly, there shall be not less than four U-bolts or twin studs per side having a 9/16-inch minimum body diameter with a 5/8-inch minimum thread diameter. Tie plates shall be at least 1/2 inch thick. The vehicle chassis frame and full length subframe shall be braced, using wood blocks at each mounting point unless the mounting point is located at a full depth crossmember. Blocks shall incorporate a keeper strap or groove for the mounting bolt and shall be of a width to assure retention. Two tieback straps shall be provided, one bolted to each side of the rear, to maintain cabinet-floor assembly alinement on the chassis frame rails or full length subframe.

3.5.2.1.2 Brackets. When brackets are used, they shall be bolted to the web of the full length subframe or to the chassis frame rails. The body mounting brackets shall provide means for drawing down the body on the subframe or chassis rails and provisions shall be made to prevent lateral shifting of the breaker strips. When additional holes are required to secure the mounting brackets to the chassis frame rails,

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they shall be located within the area of the rail which is designated as being safe for drilling in accordance with the chassis manufacturer's body builders layouts. Attachments shall not interfere with nor obstruct existing chassis components.

* 3.5.2.1.3 Breaker strips. A hardwood or dense southern yellow pine breaker strip not less than 1-1/16 inches in finished thickness shall be installed between the longitudinal sills of the cabinet-floor assembly and the vehicle full length subframe. Breaker strips shall have a taper of not less than 1/2 inch in 16 to 18 inches at the forward end.

3.5.2.2 Rigid mounting for cabinet-floor assembly. The cabinets and the loadspace floor, as an assembly or independent of one another, shall be bolted to the full length subframe. The full length subframe and the chassis frame shall be tied together in such a manner as to act as a single member resisting bending aud distortion of the cabinets. The section modulus of the full length subframe and the chassis frame combined shall be not less than 60 inches cubed per frame rail. A wood breaker strip conforming to 3.5.2.1.3 shall be provided when the cabinets and the loadspace floor are an integral assembly. When the cabinets are mounted independently of the loadspace floor, wood or metal breaker strips shall be provided when needed to provide relief at rivet heads, bolt heads, or other subframe irregularities.

3.5.3 Cabinets. Cabinet backs, partitions and shelves shall be constructed of not less than 20 gage (0.0359 inch) high tensile steel or 18 gage (0.0478 inch) carbon steel. The cabinet doors shall be of double panel reinforced construction, of not less than 0.0299 inch sheet steel. Aluminum bodies of equivalent strength may be furnished. The cabinet door jams shall be provided with neoprene tubular type gaskets, vulcanized at each corner, and shall be capable of compressing and expanding without the retention of moisture. The doors shall be weatherproof and constructed to maintain fit and alinement in service. The doors shall have a continuous rod with hem type or piano type heavy duty hinges, with stainless steel hinge pin and stainless steel, nylon, or bronze insert sockets. Latches shall have recessed handles. Cylinder type locks, keyed alike, shall be installed in each door. Moving parts of latches and locks shall be constructed of or plated with corrosion-resistant material. Cabinet doors over 44 inches in height or length shall have pillar bolts or two latches. In addition, the vertical and horizontal doors shall have a gang type drawbolt to secure the doors at the header, the sill and, for the horizontal doors, on the uprights. To secure, personnel shall turn drawbolt 90 degrees and lock with a 2-inch padlock. Horizontal doors shall have corrosion-resistant stay chains and shall be reinforced for use as shelves capable of

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supporting a live load of not less than 25 pounds on any square foot of area. The cabinets and shelves shall be capable of supporting a live load of 100 pounds on any square foot of area. Rubber fenders shall be installed around the rear wheelwells.

3.5.3.1 Cabinet arrangement. Cabinet space and dimensions shall be at least equal to that specified in table II. Cabinet arrangement shall be the manufacturer's standard and shall provide the following:

(a) Full height vertical cabinets equipped with bolted or screwed-on hooks
(b) Horizontal cabinets equipped with small parts bins with adjustable spacers, shelves and partitions and a rubber goods compartment which shall be insulated and vented
(c) Forward cabinets equipped with not less than two adjustable tray type shelves and a water cask (see 3.5.5) shall be mounted in the curbside compartment
(d) Insulated hot stick compartment
(e) Safety tread plate shall be provided on top of each cabinet
(f) Snatch block, 36 inches in length
(g) Through-box for tamper stowage with a door and recessed latch with key.

3.5.4 Lights. In addition to the lighting specified in 3.4.2.2, the following lighting and wiring shall be furnished:

(a) Two 5-3/4 inch diameter sealed beam spotlights with universal mounts shall be installed, one on each side of the cab, operable from within the cab
(b) Demountable lighting provided on hydraulic tilting pole grabber, with a service line and socket and a receptacle for lighting on the rear of the body. Lighting shall be in accordance with Federal Motor Carrier Safety Regulation 393.18
(c) Floodlight, mounted on a universal mounting, to illuminate the rear of the vehicle and a towed vehicle. The floodlight shall be provided with a toggle switch and a switch on the instrument panel
(d) An amber, rotating, vehicular warning light in accordance with MS51317-4 shall be provided. The rotating light shall be mounted at the rear of the cab on a telescoping bracket to provide clearance for aircraft loading. The lighting cable shall be of the retractable type and shall be waterproof. The rotating light shall be operated by means of a separate switch with an indicator light located on the instrument panel and marked to indicate the function of the switch

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(e) Wiring for the body shall be in accordance with the applicable portion of SAE J1292. All wiring exposed to possible physical damage, such as wiring on a cabinet floor, or wiring where tools or materials may cause the wire to be frayed or shorted shall be protected by the use of heavy wall, thermoplastic conduit or steel conduit having waterproof connections. Grommets shall be used for all wiring going through any hole or body frame work.

3.5.5 Water cask. A standard commercial, 3-gallon, insulated, potable water cask shall be furnished. The inside liner shall be stainless steel or nontoxic poly material. The outside shall be galvanized steel or equivalent. The throat area shall be of sufficient area to permit cleaning of the interior of the water cask. A brass, chrome-plated or nylon spigot, flush mounted to prevent damage, shall be furnished. Lugs and bracket(s) shall be provided to mount the cask in a vertical compartment on the curbside of the vehicle. Means shall be provided to retain the water cask and to minimize vibration during road travel and aircraft transport. A paper cup dispenser shall be furnished and installed. Means shall be provided to drain excess water to the ground.

3.5.6 Winch, rear mounted. When specified (see 6.2), a rear mounted, hydraulic planetary or hydraulic helical gear winch shall be furnished. The winch shall be mounted rearward of the turret assembly. The winch shall be located in a location accessible from the curbside access steps. The winch shall be a single drum type. The winch shall have two or three forward speeds and one reverse speed. The winch shall have a rated single line pull capacity of not less than 18,000 pounds on the bare drum. The winch shall be equipped with a hydraulic relief valve set to restrict the line pull capacity to 19,800 pounds. An integral, adjustable, oil cooled, automatic safety brake shall be furnished. The winch assembly shall be equipped with an automatic cable level wind device. A guard, as specified in 3.4.24.1, shall be furnished. The winch shall have an operating efficiency to endure 8 hours of continuous operation under load without malfunction of any component of the winch.

* 3.5.6.1 Rear winch controls. The rear winch controls shall be at the rear of the vehicle, in a location providing the operator with a clear view of the winch line as it passes through the swivel pulley and to the rear of the vehicle. The winch controls shall be located at least three feet from the swivel pulley to prevent personal injury in case the winch line should break.

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* 3.5.6.2 Rear winch line. The winch shall be wound with not less than 800 feet of 1/2-inch diameter, preformed, 6x37 improved plow steel, independent wire rope core (IWRC) wire rope and shall be equipped with a universal sheave block and hook. The winch cable shall be threaded through a tunnel beneath the floor. A swivel pulley shall be provided at the rear on the horizontal centerline of the vehicle body loadspace. The pulley shall be capable of swiveling 90 degrees to the right, left, up and down.

3.6 Turret and derrick. Turret and derrick structural components shall have a calculated design stress not to exceed 50 percent of the yield point of the materials used when the maximum rated load is applied to the center of the upper head sheave in all extended positions. For plastic components, the tensile or compressive strength shall not be exceeded with 6 times the specified load in the basket. The materials used shall meet all applicable performance requirements specified herein.

3.6.1 Hydraulic turret. The vehicle shall be furnished with a hydraulic motor driven turret. The turret shall be mounted at the rear of the cab. The turret shall be capable of continuous 360-degree rotation. All components of the turret assembly shall be capable of accepting horizontal and vertical loads specified in 3.3.3 through 3.3.8 in all positions that the derrick is capable of attaining. The turret shall have a rotating rated torque capacity of not less than 42,000 foot pounds with a safety factor of 3. The derrick shall be capable of dragging poles of a weight that shall impose on the turret the specified torque capacity of the turret (see 3.3.6). Hydraulic relief valve(s) shall be set to meet the rated loads specified herein and to provide the minimum specified torque requirements. All journals, bearings, and drive components shall be oil lubricated. A means shall be provided to rotate the turret in the event of a hydraulic or power source failure.

* 3.6.2 Derrick assembly. The derrick assembly shall be of the telescoping type, hydraulically extended and retracted by hydraulic cylinder. The derrick assembly shall be elevated from stowage position to maximum elevation by two hydraulic cylinders. The derrick assembly shall be attached to the rotating turret by replaceable, self-alining bearings and shaft. Under dynamic conditions, the derrick assembly shall support the specified load placed on the upper head sheave in all extended positions without permanent deformation of any component in or along the derrick or supporting structure. The derrick assembly shall be capable of extending and retracting in all selected positions. The derrick shall be equipped with a head sheave and a travel rack. The derrick shall be capable of setting poles up to 100 feet in length. The derrick shall have a load capacity not less than that specified in

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3.3.3. The extension and retraction shall operate by the use of one or more hydraulic control(s). All revolving components shall be furnished with lubrication fittings. All bushings shall be capable of replacement and shall be provided with lubrication fittings.

3.6.2.1 Derrick construction. The derrick shall be designed, engineered and constructed of materials that display the minimum weight conducive to the configuration and maximum performance requirements specified herein. The hydraulic lines and other associated parts which are fastened to or inside the derrick assembly shall be positioned for easy inspection, ready maintenance, or possible replacement and shall have maximum protection against damage afforded by the configuration of the derrick assembly. The derrick shall be attached to the turret by replaceable shafts. The rotating bearing surfaces of the derrick assembly at the turret which are subjected to continuous wear shall be provided with replaceable bearings or bushings. Lubrication fittings shall be provided on all rotating parts. The bearing surfaces shall be adequate to resist torsional moment when inherent with the design.

3.6.2.2 Derrick travel. The derrick travel shall be a minimum of 80 degrees to the body platform.

3.6.2.2.1 Derrick travel and extension. Derrick travel and extension shall be not less than that specified in table I.

3.6.2.3 Derrick operation. The rotating turret and derrick assembly shall be hydraulically actuated by separate mechanisms and shall operate independently and in all combinations of raising or lowering and rotating at the same time. The derrick operating mechanism shall be capable of limiting the lifting capabilities from rest position within safe limits of design stress (see 3.6) to prevent overloading. For testing purposes, means shall be provided in the hydraulic system to perform the overload tests.

3.6.2.4 Derrick storage. For normal storage, travel and shipment of the vehicle, the derrick shall be retracted and placed in an essentially horizontal position in the vertical plane parallel to the longitudinal centerline of the vehicle. Means shall be provided to hold down the derrick assembly to prevent nonactuated raising of the derrick assembly when in transit. Saddles and tiedown devices shall be in accordance with the acceleration factors specified for aircraft tiedowns (see 3.1.1.12). Derrick saddles shall be sufficiently padded to prevent damage to any component of the derrick. Saddles shall have a means to adjust the vertical height at the upper head sheave. The saddle shall have four increments of vertical adjustment. The maximum vertical adjustment shall provide a nominal vertical height of 10 feet, measured

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from ground level to the top of the vertical head sheave. Adjusting mechanism shall have positive means to prevent shear and deformation under ground transport conditions and loss of any component under any operating condition.

3.6.2.5 Capacity, height and radius indicator. An indicator shall be furnished on the operator's side of the derrick to visually display capacity, height and radius conditions under which the derrick is operating.

3.6.3 Plastic construction. The jib or insulated telescopic section, basket(s) and components shall consist of extruded, pultruded, molded or spiral wound fiberglass, and each component shall conform to the performance requirements for electrical insulation specified herein. All components shall be from a combination and multiple layers of sterile electrical grade fiberglass roving cloth or mat with high impact strength resins with weathering and fire retardant characteristics employed. The exterior surface shall be smooth and of a uniform thickness. When gelcoat is applied, the thickness of the gelcoat shall be not less than 0.006 and not more than 0.030 inch. The parts shall be free of uncured areas, gaps, cracks, holes, blisters, fuzzy surfaces, resin pockets, areas lacking resin, laps, wrinkles, delamination, air or gas pockets, patches, porosity and other similar defects. The material shall be essentially void-free. The entire surface shall be fully cured and free of tackiness. The percentage of glass by weight shall be approximately uniform. The plastic construction insulation shall meet or exceed the specified performance requirements of 3.6.3.1.

3.6.3.1 Dielectric properties. The insulated jib or insulated telescopic section shall withstand a minimum of 100,000 volts ac, 60 cycles per second applied across the full length of the jib or insulated telescopic section for not less than 3 minutes without evidence of dielectric puncture and with leakage of not more than 1.0 milliampere. All mechanisms within or along the jib or insulated telescopic section shall be made of insulating materials or so insulated to demonstrate the same dielectric properties as the jib. The insulated jib or insulated telescopic section shall be completely insulated from the derrick assembly, turret and fiberglass basket. The insulated fiberglass basket(s) shall withstand a minimum of 30,000 volts ac, 60 cycles per second, without evidence of dielectric puncture. The liner(s) shall withstand a minimum of 50,000 volts ac, 60 cycles per second, for not less than one minute, without evidence of dielectric puncture. A decal shall be furnished and installed on the basket(s) stating "NO INSULATION VALUE WITHOUT LINER."

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3.6.3.2 Nonconductive jib or section. The jib or insulated telescopic section shall be manually operated. When in place or extended, an upper head sheave height of not less than 80 feet from ground level shall be provided. The jib or insulated telescopic section shall be capable of mounting two one-man fiberglass baskets. The jib or insulated telescopic section and components shall be in accordance with 3.6, 3.6.3 and 3.6.3.1. The extension shall be marked indicating the maximum load that can be imposed on the jib or insulated telescopic section.

* 3.6.3.2.1 Jib stowage. The jib shall be hinged, pivoting back along the telescopic boom for stowage or shall be removable for stowage. Stowage provisions shall conform to 3.1.1.12. Any removable pins required shall be provided with security chains. An insulated telescopic section may be substituted for the jib. When an insulated manual telescopic section is furnished, rollers shall be provided. Wear plates alone shall not be furnished.

3.6.3.3 Aerial basket(s). Unless otherwise specified, a single, one-man, side-hung basket, having a total capacity of not less than 300 pounds, shall be furnished. The basket shall be in accordance with the performance requirements specified in 3.6, 3.6.3 and 3.6.3.1. Minimum interior dimensions of the basket shall be not less than 40 inches in depth, 21-1/2 inches in width and 21-1/2 inches front to the back. Minimum interior dimensions shall be measured at the bottom of the basket(s) just above the bottom radii. Nonrectangular baskets may be furnished provided the total usable floor area is equivalent to that of a rectangular basket with the minimum dimensions specified herein. The basket(s) shall be enclosed on four sides and the bottom, and designed to permit entrance and exit. The basket(s) and mounting bracket shall be accessible for field installation on the jib or insulated telescopic section (see 3.6.3.2) and for removal for stowage on the vehicle. When specified (see 6.2), two one-man baskets shall be furnished. When two one-man baskets are furnished, the total capacity shall be not less than 600 pounds.

3.6.3.4 Insulated basket liner(s). An insulated basket liner shall be provided for each basket. The liner(s) shall be easily removed for cleaning, and for testing the dielectric capacity. Basket liner(s) shall be in accordance with the performance requirements specified in 3.6.3.1.

3.6.3.5 Flooring. The basket(s) shall have a reinforced floor. No point in the flooring shall deflect more than 1/4 inch nor be permanently deformed when subjected to a vertical load of 300 pounds covering an area of 25 square inches. A nonconductive floor mat shall be provided for each basket.

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3.6.3.6 Basket leveling system. An independent basket leveling system shall be provided for keeping the basket horizontal regardless of derrick position. The leveling system shall consist of a basket gravity leveling system. The system shall be independently capable of keeping the basket level with the rated load on the basket with the load positioned on the floor of the basket. An adjustment shall be provided to enable personnel in the basket to control the rate of movement of the basket relative to the derrick position, and to lock the basket in a fixed position at working heights. The leveling system components shall be provided with maximum protection based on design features and safety. The system shall be insulated as specified (see 3.6.3.1). The leveling system shall operate without breakage, in all extended positions the derrick is capable of attaining.

3.6.3.7 Safety belt(s). A positive-lock type, lineman safety belt with quick-release features shall be provided for each basket. Belt(s) shall have a design strength of 2,000 pounds. The safety belt(s) shall be designed and positioned to allow freedom of movement of the workman in the basket(s). Means shall be provided to attach safety belt fasteners to the end of the jib or basket bracket and to prevent the fastener attachment from being pulled through the plastic wall, in the event of leveling system failure. Fasteners and attachments shall be corrosion-resistant.

3.6.3.8 Basket cover(s). A waterproof basket cover(s) and means to retain the cover(s) on each basket while in stowage and stowage-in-transit positions shall be provided. Provisions shall be made to secure and stow the cover(s) in a compartment of the utility body.

* 3.6.3.9 Tool trays. Fiberglass or polyethylene tool tray(s) shall be provided to fit over the edge of each basket. Provisions shall be made to secure and stow the tray(s) in a compartment of the utility body.

3.6.4 Hydraulic motor driven derrick winch. A hydraulic motor driven derrick winch shall be mounted on the turret assembly. The winch shall conform to SAE J706 and shall be equipped with an oil-cooled automatic worm brake. The winch wire rope shall be improved plow steel, 6x37, hemp core, preformed, regular lay, with cable loop formed at the end. Not less than 250 feet of 9/16-inch wire rope shall be provided. The winch shall have a rated single line pull capacity of not less than 15,000 pounds on the bare drum. The winch shall be equipped with a hydraulic relief valve set to restrict the line pull to rated capacity. The winch shall be provided with a cable tensioner that resists cable overlapping on the drum.

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* 3.6.4.1 Cable guards. Two hinged cable roller guards shall be furnished on the top of the derrick boom, one near the front and the other near the rear. The guards shall retain the cable and allow up and down movement of the cable. Rollers shall be furnished above and below the cable.

3.6.5 Outriggers. Four hydraulically operated outriggers with pivot or swivel pads shall be provided. The outriggers shall enable leveling of the vehicle on inclines and slopes and shall ensure adequate stability with relation to the ground while operating the derrick under the loads and conditions specified in 3.3.3 through 3.3.8. Outrigger cylinder hydraulic fittings, hoses, tubing and pilot check valves shall be protected by metal guards and shall be accessible for service. The guards shall withstand being stepped on without damage to the guards, hydraulic lines or fittings. Outriggers shall be designed so that they may be extended over curbings and sidewalks when operating in narrow areas.

3.6.5.1 Outrigger controls. Individual control for each outrigger shall be furnished to provide side-to-side and front-to-rear leveling of the vehicle on inclined and sloped surfaces. The right side outrigger controls shall be located on the right side near the rear, the left side controls on the left side near the rear. All outrigger controls shall be in a readily accessible location which permits operation from ground level. An automatic pilot check valve shall be provided for each outrigger to block the hydraulic fluid in the cylinder and to prevent movement of the outrigger in the event of a damaged hydraulic line. Interlocks shall be furnished to prevent the operation of the derrick assembly until all outriggers have descended and are in place on the ground surface. Interlocks shall not interfere with the operation of the hydraulic tool circuit with outriggers up or down. A self-actuating locking device shall prevent accidental raising of the outriggers while the derrick assembly is elevated from stowage position. Each outrigger shall be provided with an automatic, positive, spring actuated lock or an automatic check pilot valve which holds the outrigger in the retracted position without settling and which may be manually released.

3.6.5.2 Outrigger pads. Outriggers shall be transversely spaced not less than 16 feet, measured to the pivot or swivel of the pad. Each outrigger ground pad shall have a ground contact surface area of not less than 320 square inches. Ground pads shall be of the pivot or swivel type.

3.6.5.3 Outrigger warning light. A red warning light shall be installed in the cab in a conspicuous location to warn the driver that the outriggers are extended.

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3.6.6 Hydraulic digger. A reversible hydraulically driven post hole digger shall be provided. The motor driven digger shall be capable of boring holes up to 36 inches in diameter and 12 feet in depth. The hydraulically operated motor of the digger shall be capable of developing at the output shaft not less than 7,000 foot-pounds of torque at a shaft speed of 31 rpm. For high speed spin-off of dirt, the digger shall rotate at not less than 75 rpm. The digger shall be equipped with a boring head similar or equal to the Pengo boring heads. Hydraulic flow control valves shall be furnished to prevent overspeeding the digger motor.

* 3.6.6.1 Augers. The vehicle shall be equipped with augers to suit the boring head specified in 3.6.6. Unless otherwise specified (see 6.2), one each of 12-inch, 18-inch, 24-inch, 30-inch, and 36-inch augers shall be furnished with the vehicle. Each auger shall be furnished with a pilot bit for center-drilling the hole, and with the required number of auger teeth. Auger teeth shall be capable of boring holes in swamp, loose sand, hard terrain, and rocky soil. One pilot bit extension shall be furnished. One universal mount shall be provided on the truck, capable of holding any one of the augers, at the operator's option. Stowage space for the 36-inch auger shall be capable of accepting the air transport tiedown requirements specified herein (see 3.1.1.12). Temporary tiedowns suitable for holding all augers for initial shipment to destination shall be provided.

3.6.6.1.1 Auger teeth. One complete set of spare auger teeth shall be furnished for each auger furnished. One spare pilot bit extension shall be furnished.

3.6.6.1.2 Digger and auger storage on derrick assembly. Stowage and a hydraulically operated mechanism for stowing the digger, with up to a 24-inch size auger in place, shall be provided on the derrick assembly. The stowage mechanism shall raise and place the digger and auger into a parallel position alongside the derrick assembly and shall lock the digger and auger automatically with a positive lock. The stowage mechanism shall provide for hydraulic release of the digger and auger from the derrick control station. A slip clutch or other torque limiting device shall be provided for the winding cable. A safety check valve shall be furnished to prevent the clutch and digger from slipping. Stowage provisions shall have anti-bounce provisions for the digger and the auger.

3.6.6.2 Hydraulic pole grabber. A hydraulically tilted pole grabber shall be furnished. The pole grabber shall be located at the end of the derrick assembly. The assembly shall be capable of setting poles up to 100 feet in length. The pole grabber assembly shall consist

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of two steel arms actuated by a hydraulic cylinder. The pole grabber shall be provided with an automatic safety pilot check valve. The assembly shall be capable of tilting not less than 80 degrees.

* 3.6.7 Controls. A complete set of hand-operated controls for the hydraulic turret, derrick assembly, winch(es), digger and pole grabber shall be provided. Controls shall be single-lever or multi-lever type. Controls shall be furnished on a control console on the rotating turret. An operator's seat shall be furnished. At least the following gages and controls shall be furnished on the console:

(a) Key-operated start and stop switch
(b) Foot operated engine throttle control
(c) Tachometer
(d) Console dash light(s) with rotating shield(s) and toggle switch to illuminate console panel
(e) Pressure and temperature gages for the hydraulic system
(f) Restricted return line filter warning light.

All gages, controls and switches furnished shall be mounted to resist shock and shall be weatherproof. Multi-lever controls shall be spaced to allow feathering and the operation of 2 or more functions at once. Smooth operation, with no lurching of the derrick assembly, shall be possible. The controls shall be self-centering and shall revert to a neutral position when released by the operator. A safety interlock system shall be provided to prevent operation of the derrick assembly when the outriggers are not extended. A means shall be provided to retract and lower the derrick assembly and rotate the turret into the stowage position, in the event of a hydraulic failure. The switches, instruments, and pressure gages on the lower control station shall be panel mounted and identified as to their function and installed in a manner to facilitate removal and service. Gages and instruments shall be shockproof and weatherproof. Identification markings shall be on a photoetched, corrosion-resistant plate.

3.6.7.1 Control console. The control console specified in 3.6.7 shall be weatherproof. An operator's platform shall be furnished with safety tread steel plate or grating. A weatherproof cover shall be provided to cover the controls. Means shall be provided to retain and secure the cover.

3.6.7.2 Overhead guard. A safety guard shall be furnished to provide protection for the operator. A minimum of 39 inches shall be provided from the top of the seat to the underside of the guard. The guard shall not interfere with the operation of the truck and equipment.

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The guard shall be foldable or removable for air transportability. The guard shall be of see-through design and shall withstand the force of a falling object, weighing not less than 25 pounds, from a height of not less than 80 feet, without penetration or total collapse.

* 3.6.7.3 Controls in basket. In addition to the lower controls, controls shall be provided at the basket for rotating, extending, retracting, elevating and lowering the derrick to its fullest extent. Controls shall be designed to prevent inadvertent actuation by the operator. The lower controls shall be capable of overriding the basket controls and of rendering the top controls inoperative. Basket controls shall be of the fiber optic type with proportional full range and metering capability conforming to the requirements specified herein. When two baskets are furnished, the controls in the basket shall be within easy reach of either basket. The controls at the basket shall be located at the right or left side of the basket in a position easily accessible to the operator facing the work area. Controls on the front or rear of the basket shall not be furnished. The controls shall be identified as to their function. When two baskets are furnished, the identification plates shall be located to be seen from either basket. The identification plates shall be located at the controls and shall be readable in all positions of the extended derrick.

* 3.6.7.4 Fiber optic cable (link) control system. The fiber optic control system shall be of the digitalized, multiplexed type, and shall provide remote control and complete electrical isolation between the controls in the basket and the vehicle. The system shall include proportional input controllers and on-off switches; battery powered transmitter processor; a fiber optic cable (link) connecting the transmitter and receiver; an electronic receiver; and four proportional electro-hydraulic control valves. The receiver shall provide cable termination for the vehicle battery connections and output terminals to the electro-valves. The fiber optic cable (link) shall be permanently installed. The transmitter shall be capable of removal and installation without detaching or exposing the fiber optic cable (link). The system shall function properly in ambient temperatures from minus 40°F to plus 145°F. The system shall be capable of remote control operation from a 100-foot minimum radius out from the control console on the rotating turret. The system shall not malfunction, either temporarily or permanently, when subjected to common electromagnetic sources and the test procedures for radiated susceptibility as stated in SAE J1113. The control system shall be shielded to ensure proper operation and nonsusceptibility to fields encountered in use in proximity to electrical transmission lines up to 100 kilovolts alternating current (kVac). The receiver and transmitter shall be in enclosures of weatherproof construction meeting the requirements of NEMA 12. The

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enclosures shall be impervious to dust and dirt contamination, and shall be designed to withstand an impact force of not less than eleven times the force of gravity without malfunction or damage to the components.

* 3.6.7.4.1 Controller. Proportional type controllers for all operations shall be furnished. The controllers shall be of heavy duty construction designed for mobile industrial equipment and shall have a design operating life of not less than one million cycles. Controller design shall provide for easy replacement of electronic subassemblies in the field. Subassemblies shall include, but shall not be limited to, potentiometers, electronic circuit boards, and microswitches. The controllers shall be of weatherproof construction and shall include nonholding detents for fast and slow operation. Controls shall have a mechanical or deadman electrical interlock to prevent inadvertent actuation. Signals generated by controller handle movement shall be conditioned by the adjustments at the transmitter. Adjustments shall include a current threshold adjustment, to eliminate nonlinear operation in valve response at the start of handle movement; a maximum output current limit adjustment; and a ramp adjustment to provide slow speedup and slowdown to the function being controlled.

* 3.6.7.4.2 Transmitter. The transmitter shall be in an enclosure located in the basket and shall be capable of accepting signals from the controllers and on-off switches. The transmitter shall be capable of converting the signals to coded pulse outputs which shall be used to drive a near-infrared light emitting diode (LED). Each coded pulse output to the receiver shall contain not less than three different parameters to detect any occurence of errors in transmission. Errors in transmission shall include, but shall not be limited to, framing error detection, sync detection, and parity error detection. In addition, the transmitter shall produce two separate coded outputs to each proportional electrohydraulic valve. The transmitter shall have an LED type indicator to display transmission status. The transmitter enclosure outside dimensions shall be not more than 16 inches by 15 inches by 6 inches and shall weigh not more than 25 pounds.

* 3.6.7.4.3 Receiver. The receiver shall be in an enclosure located on the vehicle and shall be capable of accepting signals from the transmitter through the fiber optic cable. The receiver input voltage range shall be not less than 10 volts dc and not more than 18 volts dc. A power conditioning circuit shall be provided. The circuit shall supply sufficient voltage to operate the receiver and starting relay under low battery voltage conditions, and when starting causes battery voltage to drop to not less than 4 volts. The receiver shall have three LED type indicators to display: "PROPER POWER APPLIED"; "LINK/DATA O.K."; and "OUTPUT POWER STATUS." The receiver shall include a

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photosensitive device, a coded pulse decoder, an error detection circuit, and an amplifier. The photodetector shall convert near infrared light pulses from the transmitter into electrical pulses of the same duration. The output from the photodetector shall be fed to the receiver-decoder. The decoder shall be capable of accepting digital data and converting it to parallel binary data. Data determined valid shall be converted to an analog signal and shall be supplied to an output drive channel which shall provide adequate power to drive the proportional electrohydraulic valves and control solenoids. Any invalid data, loss of signal, or any other malfunction shall cause an electro-mechanical disconnect of power to all outputs. Current regulation proportional to controller actuation shall be provided to ensure accurate and continual valve actuation. The output current shall not vary more than ± three percent for the temperature range specified (see 3.6.7.4), for the voltage range specified herein, and for a valve resistance range of ± 25 percent of the valve electrical resistance. The receiver enclosure outside dimensions shall be not more than 14 inches by 13 inches by 7 inches. The receiver shall be equipped with a selector switch labeled "MANUAL/BUCKET." The selector switch shall override control of the transmitter.

* 3.6.7.4.4 Fiber optic cable (link). The fiber optic cable shall be a low loss, single core, silica glass or plastic fiber optical cable with a dielectric capacity of not less than 100,000 volts per meter. All cable components shall be nonhygroscopic. The cable shall be capable of repeated bending over a radius of not less than 1-1/2 inches without damage. A polyurethane jacket shall provide mechanical protection. Stainless steel end fittings shall be furnished for positive connection of the cable to the transmitter and receiver. The cable shall be run inside a hose for additional protection. The cable shall be installed free of kinks and radii of less than 1-1/2 inches.

* 3.6.7.4.5 Hydraulic dump valve. A hydraulic dump valve shall be provided. The dump valve shall be installed in the hydraulic line between the hydraulic pump and the electrohydraulic valves. The dump valve shall be a 3-way valve and shall be designed to direct fluid to the electrohydraulic valves when the receiver is energized. In the event of receiver power interruption, the dump valve shall de-energize and direct hydraulic fluid to the reservoir.

* 3.6.7.4.6 Rechargeable battery packs. A rechargeable battery pack to energize the transmitter shall be furnished. A spare battery pack shall be furnished. The battery packs shall be capable of not less than 48 hours of continuous operation at a voltage range of 8 to 12 volts. Both battery packs shall be of the sealed nonspillable type with mounting case and quick disconnects.

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* 3.6.7.4.7 Battery charger. A battery charger shall be furnished to charge the battery pack not in use. The battery charger shall be permanently mounted in a readily accessible location within the vehicle cab. The charger shall include a push-to-test switch, a red LED indicating low charge status, a green LED indicating charged status, and an indicator to indicate that the battery is charging. The charger shall be capable of either a quick recharge of not more than 2 amps or a float charge of not more than 20 milliamps. The charger shall function throughout the temperature range specified in 3.6.7.4. The charger shall be connected to the vehicle electrical system.

3.6.8 Hydraulic system. The hydraulic system shall consist of the components required to perform all hydraulically actuated operations of the derrick assembly, winches and outriggers with the degree of control specified herein. Mechanisms and reservoirs shall be positioned or shielded to prevent hazardous fluid contact with heated surfaces or electrical apparatus. An electric push-button throttle control shall be furnished and located on the right rear corner of the body. The electric push-button shall activate the engine throttle to increase the engine speed to that required for operation of the outriggers, pole puller and tamper, and to return the engine to idle speed. An accumulator may be used as a component of the hydraulic system to provide the maximum degree of temperature control under continuous service.

* 3.6.8.1 Hydraulic cylinders. Hydraulic cylinders shall be double acting. All hydraulic cylinders shall incorporate seals to prevent entrance of moisture and dirt into the cylinders and leaks of hydraulic fluid from the cylinders. Seals applied to reciprocating parts shall be of a design and type proven in hydraulic applications. All hydraulic cylinders for actuating the derrick and outriggers shall be provided with pilot check or holding valves to prevent descent of the derrick assembly, and retraction of the outriggers because of hydraulic pressure loss. Means shall be provided to install pressure gages for testing hydraulic system pressure and relief settings. All surfaces of hydraulic ram piston rods shall be stainless steel or chrome-plated with sufficient thickness to withstand a 30-hour salt spray test in accordance with ASTM B117.

* 3.6.8.2 Hydraulic lines and fittings. Hydraulic lines shall be located to facilitate maximum ease of removal and installation, visual inspection for failures, and maximum protection from damage during operation of the derrick assembly. Flexible hydraulic hose or collector rings shall be used where flexing of hydraulic lines is encountered and where installation of components requires a flexible coupling. Standard straight-thread, flared type fittings or adaptions to flared type fittings, or tapered pipe thread with dry seal fits shall be furnished.

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Hydraulic lines to the digger shall be of the telescoping type. The hydraulic lines and fittings shall be air- and liquid-tight and shall be installed to prevent vibration and chafing. The system shall be designed to prevent hydraulic transients. Provisions shall be made for bleeding air from the hydraulic system and completely draining all hydraulic fluid lines. A manual type bleeder shall also be provided as on-vehicle-equipment on closed-center systems. The hose assemblies and all seamless steel hydraulic lines shall be adequate to provide the rate of flow, temperature and pressures required to operate the system smoothly and efficiently. Galvanized pipe and copper tubing shall not be furnished. A pressure gage shall be provided on the discharge side of the hydraulic pump. When a multi-section pump is furnished, a pressure gage for each section of the pump shall be furnished.

3.6.8.3 Hydraulic pump system. A hydraulic pump with necessary pressure and flow capacities to operate all hydraulic powered components shall be provided. The hydraulic pump shall be driven by a full torque, heavy duty power takeoff unit attached to the truck chassis transmission. Chain drives shall not be used. The pumping system shall include an adjustable pressure relief valve or unloading valve with the range of adjustment which allows raising or lowering of the derrick assembly with any working or test loads specified herein.

3.6.8.4 Hydraulic fluid reservoir. A hydraulic fluid reservoir shall be provided and shall have sufficient capacity to hold the required volume of hydraulic fluid, with not less than 10 percent for expansion space. The reservoir shall be fully baffled. The reservoir shall be provided with a filler opening and removable oil strainer and a vented filler cap, accessibly located to facilitate filling the tank from a one-gallon container using a funnel. The vented filler cap shall have means to prevent fluid from splashing out of the tank during movement of the vehicle. The filler cap shall incorporate an air filter and shall be attached to the tank by a corrosion-resistant chain. Provision shall be made for inspection and cleaning of the interior and draining of the reservoir. The reservoir drain shall be located to prevent fluid from draining on other components of the vehicle and to allow recovery of the fluid in a container. The reservoir or the return and discharge lines to the reservoir shall be provided with readily accessible 1-inch nominal diameter quick disconnect fittings arranged for attaching an auxiliary filler and pump for cleaning the hydraulic fluid in the reservoir. The reservoir shall have a photoetched, corrosion-resistant caution plate indicating the type of hydraulic fluid required and the maximum fluid level in the reservoir with the derrick in the travel position and the hydraulic lines and components full of fluid. A sight level or dipstick gage shall be furnished to indicate the level of the fluid. The tank shall be clean.

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3.6.8.4.1 Hydraulic fluid. Unless otherwise specified, hydraulic fluid conforming to the manufacturer's standard or recommended requirements shall be provided in the hydraulic system. When specified (see 6.2), hydraulic fluid conforming to MIL-H-83282 shall be provided in the hydraulic system. A permanent decal located near the hydraulic system fill port shall be furnished stating "CAUTION: USE HYDRAULIC FLUID CONFORMING TO MIL-H-83282 ONLY."

* 3.6.8.5 Oil filter and strainers. A full flow type 10-micron discharge oil filter in the return line to the reservoir shall be provided. A 100-mesh suction liner strainer in the hydraulic reservoir shall be provided. The filter and the suction strainer shall be easily accessible for periodic cleaning, or for replacement of the filter cartridge. The filter and the strainer shall have sufficient capacity to allow the pump to deliver its maximum rated flow. The return line filter shall be equipped with a restriction sensing device. The sensing device shall be wired to a red light at the control console marked "FILTER RESTRICTION."

3.6.8.6 Hydraulic fluid temperature. The hydraulic fluid temperature shall be maintained within safe limits to prevent seal damage or approach to flash point of the fluid during all operating conditions of the vehicle and extended use of the derrick assembly. Excessive heating (more than 55°F above ambient, in an ambient of 125°F, or equivalent in lower ambients) of the fluid shall not be acceptable under any operating condition. Provisions shall be made to prevent malfunction or unactuated operation of the hydraulic control system due to changes in fluid temperature.

3.6.9 Ground hydraulic tools.

3.6.9.1 Ground hydraulic tool outlet. A hydraulic tool outlet shall be provided on the rear or on the curbside in a protected position. Discharge port(s) and return port(s) shall be provided with quick-coupling connectors with corrosion-resistant caps, secured with corrosion-resistant chains to protect the connectors and the valve from contamination. Flow divider control valves shall use the hydraulic fluid from the hydraulic system of the vehicle.

3.6.9.1.1 Hydraulic tool circuit. The hydraulic tool circuit shall be provided with quick disconnect, dripless couplers for attaching the pressure and return hose lines for the motor driven power tools.

3.6.9.1.2 Hydraulic fluid flow capacity. Unless otherwise recommended by tool manufacturer(s), the hydraulic fluid flow capacity shall be not less than 3.8 and not more than 4.2 gallons per minute

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(gpm) at a pressure of not less than 1,700 and not more than 1,800 pounds per square inch (psi). The back pressure in the return line shall not exceed 150 psi.

3.6.9.2 Ground tool hose. Two 50-foot lengths of hydraulic, wire braid hose with quick disconnect, dripless couplers on both ends with caps for use on the ground level shall be furnished. Marking shall be provided on the instruction plate (see 3.1.1.1.1), that the wire braid hydraulic hoses shall not be used on or near electrical lines. The hoses shall be furnished with fittings compatible with the tool outlets specified in 3.6.9.1. Hoses shall be secured in accordance with 3.1.1.12.1.

3.6.9.3 Hydraulic pole puller and jack. A hydraulic pole puller and jack shall be furnished. The pole puller shall have the capacity to lift a pole 100 feet in length, when the pole is in the ground to a depth of 12 feet. The pole puller shall be provided with a chain, a base plate and hydraulic controls. The pole puller and jack shall have a lifting force of not less than 40,000 pounds and shall be capable of vertically lifting the pole not less than 15 inches per cycle. The jack shall be operated from the ground tool outlet specified in 3.6.9.1 using the hoses specified in 3.6.9.2. Means shall be provided to secure the pole puller and jack in accordance with 3.1.1.12.1.

3.6.9.4 Hydraulic tamper. A hydraulically operated tamper shall be furnished. The tamper shall be capable of tamping the earth around the 100-foot pole, when the pole is 12 feet in the ground. The tamper shall be capable of delivering a force of not less than 25 pounds at a rate of not less than 1,200 impacts per minute. Hydraulic control shall be furnished. The tamper shall be operated from the ground tool outlet specified in 3.6.9.1 using the hoses specified in 3.6.9.2. The tamper shall be secured as specified in 3.1.1.12.1.

3.6.9.5 Hydraulic capstan hoist. A hydraulic capstan hoist shall be furnished. The hoist shall be equipped with quick disconnect, dripless fittings and shall utilize the hoses specified in 3.6.9.2 and hydraulic power from the tool outlet (see 3.6.9.1). The drum shall be one direction drive and shall have a backstop on the output shaft to prevent reverse travel. Lowering will be accomplished by letting the rope slip on the drum. The capstan hoist shall be complete with a link chain and screw clamp attachment device for mounting to surfaces other than a round pole. The hoist shall have a rated single line pull capacity of not less than 1,250 pounds on the bare drum. The hoist shall have a remote control, foot operated, open-center valve. The capstan shall utilize the hose specified in 3.6.9.2. The capstan shall be secured as specified in 3.1.1.12.1.

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3.6.10 Basket hydraulic tools.

3.6.10.1 Basket outlet and hose reel. When specified (see 6.2), a hydraulic tool outlet shall be furnished at the basket(s) by means of a hose and a hose reel or other automatic hose stowing arrangement. The hose reel shall be mounted in an accessible location on the turret or the derrick. The hose shall automatically extend and retract with the movement of the derrick. The hose and reel shall be capable of withstanding 1,800 psi line pressure and meeting the dielectric properties specified in 3.6.3.1. The hose shall have quick disconnect, dripless fittings at the basket outlet. A weatherproof cover shall be provided for the hose reel. The following shall be included:

(a) Flow control valve
(b) Line check valve
(c) Roller hose guides.

* 3.6.10.1.1 Basket tool hose. When a basket hydraulic tool outlet is specified, an 8-foot dual hydraulic hose, for use in the basket, shall be furnished. The hose shall be nonconductive type rubber or nylon with compatible quick disconnect, dripless fittings on each end. The hose ends shall be furnished with dust caps.

3.6.10.2 Basket hydraulic chain saw. When specified (see 6.2), a hydraulic chain saw shall be furnished. The saw shall be equipped with quick disconnect, dripless fittings. The saw shall have a chipper type chain and shall be suitable for cutting trees up to 15 inches in diameter. A suitable stowage area in the body shall be provided (see 3.1.1.14.1).

3.6.10.3 Basket hydraulic chain saw, pole mounted. When specified (see 6.2), a hydraulic chain saw, pole mounted, shall be furnished. The chain saw shall be lightweight (approximately 10 pounds) and have a cutting capacity of 9 inches. The saw shall be equipped with quick disconnect, dripless fittings. The pole and saw assembly shall be approximately 8 feet long and shall have a fiberglass type of body to provide good insulation. A suitable stowage area in the body shall be provided (see 3.1.1.12.1).

* 3.6.10.4 Basket hydraulic circular saw, pole mounted. When specified (see 6.2), a hydraulic circular saw, pole mounted, shall be furnished. The saw shall be equipped with quick disconnect, dripless fittings. The saw shall have a capacity of not less than 3-1/4 inches and shall operate at approximately 8,900 feet per minute. The pole and saw assembly shall be approximately 6-1/2 to 7 feet long and shall have a fiberglass type of body to provide good insulation. Two saw blades

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shall be furnished, one with not less than 40 teeth and one with not less than 80 teeth. Suitable stowage area in the body shall be provided (see 3.1.1.12.1).

3.6.10.5 Basket hydraulic impact wrench and drill. When specified (see 6.2), a hydraulic impact wrench and drill shall be furnished. This combination tool shall be reversible and capable of installing or removing lag screws or other hardware. The wrench shall have a 5/8 standard bolt size capacity with a 1/2-inch square drive. The tool shall be capable of developing not less than 200 foot pounds of torque with at least 1,350 impacts per minute and shall be equipped with a torque regulating device. A quick change drill chuck shall be furnished. Quick disconnect, dripless fittings shall be provided. Suitable stowage area in the body shall be provided (see 3.1.1.12.1).

3.6.10.6 Basket hydraulic pruner, pole mounted. When specified (see 6.2), a hydraulic pruner, pole mounted, shall be furnished. The pruner shall be equipped with quick disconnect, dripless fittings. The pruner shall have a capacity of 2 inches in diameter. The pruner shall be mounted on a fiberglass type pole to provide insulation and shall be approximately 7 feet long. Suitable storage area in the body shall be provided (see 3.1.1.12.1).

3.6.11 Telephone maintenance accessories. The following equipment shall be furnished:

(a) Capstan for the winch drive shaft extension (AT&T type), on the front mounted winch
(b) Collapsible cable reel attachment for front mounted winch
(c) Swivel hook with downhaul weight for derrick winch.

3.6.12 Safety. All equipment or exposed portions of the equipment which are subject to extreme temperatures and inclement weather, and all rotating or reciprocating parts which are of such nature or so located as to become a hazard to operating personnel, shall be insulated, fully enclosed, or properly guarded.

3.7 Dissimilar metals. All dissimilar metals used throughout the vehicle shall be insulated from one another to prevent galvanic or electrolytic action.

3.8 Lubrication system. A lubrication system shall be furnished with fittings for all bearings and other working parts which require lubrication for continued service. The vehicles shall be lubricated prior to delivery with the applicable grade of standard lubricant for the ambient air temperature at the delivery point.

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3.9 Stress analysis. A stress analysis shall be performed. The stress analysis shall include the weight of all major items permanently installed on or within the vehicle, and shall show the location of the center of gravity of such component relative to the vehicle axles (travel and critical working positions). In addition, the analysis shall include the value of stresses of the turret and at the base of the derrick; at the area between all upper and the lower derrick sections; at the area between the uppermost derrick section and plastic jib or insulated telescopic section; at the hinge area between the plastic jib or insulated telescopic section and the basket; and the center of gravity of the lower and all upper derrick sections and plastic jib or insulated telescopic section. The analysis shall verify that the stress limits specified in 3.6 are is not exceeded under the conditions specified in 3.6.

3.10 Servicing and adjusting. Prior to acceptance of each vehicle by the Government inspector, the contractor shall service and adjust each vehicle and its mounted equipment for operational use including at least the following: cycling the entire turret and derrick hydraulic system; alinement of lights; adjustment of engine and brake systems; filling and charging of batteries; alinement of front wheels; inflation of all tires; complete servicing of chassis, engine, running gear and mounted equipment with grades of lubricants and hydraulic fluids recommended for the ambient air temperature at the delivery point; servicing of the cooling system with a solution of ethylene glycol type antifreeze and water in equal parts by volume; and servicing of the windshield washer reservoir with water and appropriate additives.

3.11 Workmanship. The quality and workmanship of all component parts, including fabricated parts, machined parts, and castings shall be commensurate with sound manufacturing standards for similar equipment. The vehicles shall be free of all burrs, sharp edges, die, tool and metal handling marks, rust, scale and cracks. All metal and plastic materials used shall be free of delamination defects, and shall be prime material. Defective components, or parts and assemblies which have been modified to overcome deficiencies, shall not be furnished.

3.11.1 Welding. Adequate preparation of metal prior to welding of the body turret, digger and components shall be provided. Surface of parts to be welded shall be free of rust, scale, paint, grease and other foreign matter. Welding rods shall be compatible with the metal being welded. All critical arc welds shall be magnetic particle or dye penetrant inspected. The following defects shall not be acceptable:

(a) Welds which are ground flush with metal
(b) Slug or plug welding on load bearing members

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(c) Undersize of fillet welds, overlaps, inadequate fusion and penetration, undercutting of weldments and vertical down welding
(d) Unremoved slag and spatter
(e) Cracks
(f) Spot, tack or intermittent welds for strength.

Weld penetration shall be such as to provide transference of maximum design stress through the base metal juncture. Intermittent welding shall be permissible when in accordance with 3.11.2 and 3.11.3.

3.11.2 Workmanship specimens. Workmanship specimens shall be prepared by the manufacturer for the subframe and derrick components to simulate all welding conditions. The specimens shall represent the minimum acceptable weld quality and cleaning procedure to be used in production. Upon submittal of these specimens by the contractor, the quality level and identification shall be subject to approval by the Government. The specimens shall be prepared using the welding procedures to be employed in production. Specimens prepared to represent multiple pass welds shall be made in such a manner as to leave at least two inches of each layer of weld metal exposed. Approval specimens shall be retained at the contractor's plant.

3.11.3 Welding procedures. The welding procedures shall be recorded prior to production of any weldment, but no procedure qualification other than workmanship specimens shall be required.

3.11.3.1 Factors. The factors listed below which are applicable shall be included in the recorded welding procedures:

(a) Joint sketch
(b) Joint type and preparation
(c) Base metal type
(d) Filler metal, size and type
(e) Position of welding
(f) Arc-voltage, current range and polarity
(g) Manual or machine
(h) Travel speed, machine only
(i) Pre-heat and post-heat.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other

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facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility for assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Government verification. Quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of observation of the operations to determine that practices, methods and procedures of the contractor's inspection are being properly applied. Failure of the contractor to promptly correct product deficiencies discovered shall be cause for suspension of acceptance until correction has been made or until conformance of product to specification criteria has been demonstrated.

4.3 First article or first production vehicle inspection. The first vehicle produced under the contract, whether a first article or the first production vehicle (see 3.1), shall be inspected by the contractor at his plant under the direction and in the presence of Government representatives. The purpose of the inspection shall be to determine vehicle conformance to the contract. The first vehicle shall be representative of the remaining vehicles proposed to be furnished under the contract. Acceptance of the first vehicle shall not constitute a waiver by the Government of its rights under the provisions of the contract. Failure to pass the following tests shall be cause for rejection. Any malfunction, excessive maintenance, permanent deformation, failure, tire scuffing, or overheating shall be cause for rejection.

* 4.3.1 Vehicle weight. The first vehicle shall be weighed to determine the curb weight and distribution of curb weight on the front and rear axles. The total imposed loading on the front and rear axles shall be computed by the contractor and verified by the Government, using the curb weight, the operator weight at 175 pounds and the payload

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required to provide the specified GVW. The calculated imposed loads on the front and rear axles shall be compared to the suspension, axle and tire load capacity ratings to determine if these components are of adequate capacity to meet contractual requirements.

4.3.2 Road test. The first vehicle shall be examined and road tested by the contractor to assure that the vehicle will operate in accordance with contractual requirements. The vehicle, loaded with payload and with all equipment specified herein, shall be driven at speeds up to at least 50 mph for a minimum distance of 50 miles. The distance traveled shall consist of highway, rough dirt roads, partly surfaced gravel roads, puddles, and sharp turns. For each 10 miles of travel at least one sudden stop shall be accomplished. Upon completion of the road test, all equipment shall be thoroughly examined. The equipment shall show no evidence of damage; all compartment and cab doors shall properly open and shut and there shall be no evidence of mud or dust inside; and there shall be no evidence of racking of the body.

* 4.3.2.1 Air transportability physical verification. The dimensions of the first vehicle shall be compared to the engineering drawings upon which air transportability approval was granted for verification of overall size, angle of approach and angle of departure, ramp crest clearance and other details. The first vehicle shall be weighed in accordance with 4.3.1, and the actual weights shall be compared to the air transportability calculation figures.

4.3.3 Truck body treatment and painting. A certification regarding the body cleaning, treating, prime painting and salt spray resistance testing required by MIL-STD-1223 shall be made to Government representatives at the first production vehicle inspection.

4.3.5 Height, reach, and load and rotational speed tests. The vehicle shall be placed in a clear, level area, providing for unrestricted movement of the derrick. A tape measure or other measuring device shall be attached at the centerline of the upper head sheave and the derrick placed in each of the positions specified in table I. The sheave height and the angle of the derrick to the horizontal shall be measured to determine conformance to 3.3.3. During the tests, the derrick assembly shall be subject to, but not limited to, the following checks:

(a) Check to determine whether the derrick assembly can be elevated without outriggers being extended
(b) Determine if outriggers are locked automatically when in an extended position and in a retracted position
(c) Operate equipment using each set of controls separately

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(d) Check to insure that the entire system can be operated smoothly and without jerk motions in all positions
(e) Check the hydraulic system, including the pump, at 10 percent above the pump manufacturer's maximum operating pressure setting for normal workload conditions.

4.3.5.1 Load tests. The derrick shall be extended and with a measured load of not less than 1,000 pounds suspended from the derrick winch line, the derrick shall be rotated, retracted, extended, elevated, lowered, stopped, restarted and the winch line raised and lowered, all in a random manner and in random combinations, at full speed, through all positions the derrick is capable of assuming. One timed revolution at maximum speed shall be made. The load shall be changed to 7,000 pounds and the tests repeated, except the derrick shall not be lowered to less than 80 degrees above horizontal. Stalling of the vehicle; inability to rotate, elevate, lower, extend, or retract into required positions at required speeds; lifting of any vehicle wheel off the ground; part failure or permanent set shall constitute failure of the test.

4.3.5.2 Tests with fiberglass jib. With the fiberglass jib installed or the insulated telescopic section extended, measurements from the sheave on the fiberglass jib or insulated extension and from the bottom of the aerial baskets or insulated telescopic extensions, both to ground, shall be made to determine conformance of the jib or insulated telescopic section and baskets to 3.3.3. With the derrick extended, the fiberglass jib in place or the insulated telescopic section extended, and twice the load specified in 3.6.3.3 placed in the baskets, the derrick shall be operated through all attainable positions. With the engine shut off, the baskets shall be lowered from maximum reach position to not more than 45 inches above ground level, retracting the telescopic sections the minimum required to bypass the limiting devices called for in table I, footnotes 1/ and 2/. The test shall be repeated, without baskets, with 1,500 pounds suspended over the fiberglass jib head sheave or the extended telescopic section head sheave, except the derrick shall not be lowered to less than 80 degrees above horizontal. Stalling of the vehicle; inability to rotate, elevate, lower, extend or retract into required positions; failure of the baskets to stay level; failure to lower at a safe rate with the engine shut off; lifting of any vehicle wheel off the ground; part failure or permanent set shall constitute failure of the test.

4.3.5.3 Horizontal pull test. The derrick shall be placed in a horizontal position at right angles to the longitudinal centerline of the vehicle. With the derrick fully retracted and the winch line over the upper head sheave, a tension scale or dynamometer shall be placed in

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the winch line, parallel to the longitudinal centerline of the vehicle and attached to a fixed object. The derrick shall be rotated away from the fixed object and the winch line reeved in to take up slack and maintain the derrick at right angles to the vehicle, until the horizontal component of the force reading on the scale or dynamometer at a right angle to the derrick, produces a torque of not less than 42,000 foot-pounds. The horizontal component of the force is the scale or dynamometer reading times the cosine of the angle between the winch line and the horizontal. Stalling of the vehicle; malfunction; evidence of part failure or permanent set; operation of the turret hydraulic relief valve prior to reaching the required torques; or failure of the hydraulic relief valve to operate prior to overload shall constitute failure of the test.

4.3.5.4 Static tests. The vehicle shall be secured to the ground and the winch line rigged to permit operation of the derrick with overloads. Using a tension scale or dynamometer and a ground anchor, or measured weights, the derrick shall be subjected to twice the loads specified in 4.3.5.1 and 4.3.5.2 (without baskets). The derrick, without the jib or fiberglass extension, shall be extended and placed in its most vertical position. A load of 14,000 pounds shall be suspended from the upper head sheave. Part failure or permanent set shall constitute failure of the test.

4.3.5.5 Detection of failure. After completion of the tests specified in 4.3.2 through 4.3.5.4, dye penetrant or magnetic particle inspection shall be applied on the critical areas of welds. A water soluble penetrant shall be applied on all critical plastic areas. All such inspections shall be conducted by a technican qualified and experienced in these procedures. The water soluble penetrant or magnaflux shall provide a means to inspect for crack indications after dynamic and static conditions. Any evidence of cracks or permanent deformation shall be cause for rejection of the vehicle. At the manufacturer's option, dye penetrant or magnetic particle inspection may be conducted prior to the first vehicle inspection provided all tests specified in 4.3.2 through 4.3.5.4 are conducted prior to the inspection; records of these tests are presented at the first vehicle inspection; and records of the tests and dye penetrant or magnetic particle inspection witnessed by the Government are presented at the time of the first vehicle inspection.

4.3.5.5.1 Critical areas. At least the following areas shall be inspected by magnetic particle or dye penetrant:

(a) Outriggers and attaching areas
(b) Turret base, trunnion, and yoke at attachment of the derrick and struts

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(c) Cylinder clevis brackets
(d) Lower sheaves, sheave supporting brackets, cable anchor and support brackets
(e) Upper sheaves, sheave supporting brackets, cable anchor and support brackets
(f) Structural steel of upper and lower derrick sections
(g) Upper derrick and yoke at attachment to basket and side brackets
(h) Fiberglass jib or telescopic section (dye penetrant).

4.3.6 Position holding ability. The vehicle shall be placed in a longitudinal position on a grade of not less than 15 percent and not leveled. The telescoping section position, height, angle, and load producing maximum tipping moment within the conditions specified in table I shall be calculated. The derrick shall be placed under these conditions and rotated up and down the grade to determine conformance to 3.3.4 and 3.3.5. The derrick shall be placed transverse to the longitudinal centerline of the vehicle and the vehicle engine shut off. The position of the upper head sheave relative to a fixed point shall be noted and recorded. The position of the upper head sheave relative to the same fixed point shall be noted after an elapsed period of not less than 1 hour to determine conformance to 3.3.7.

* 4.3.7 Temperature stabilization test. To determine conformance to 3.6.8.6, the vehicle shall be tested. The vehicle shall be placed in an area suitable for digging holes in firm, undisturbed soil and prepared for digger operation. The auger used shall be of the 36-inch size, except that for frozen ground or other extremely hard soil, pilot holes using a smaller auger may be dug first, prior to digging with the 36 inch auger. The ambient temperature shall be measured and a thermocouple or thermometer inserted in the hydraulic fluid reservoir at its inlet from the oil return line or, if an oil cooler is provided, at the inlet to the oil cooler. The digger shall be suspended just above the ground and continuously rotated at not less than 75 rpm. The temperature of the hydraulic oil at the hydraulic oil return line or at the inlet to the oil cooler, as applicable, shall be observed and the digger rotated for one hour or until this temperature has stabilized, whichever is less. A temperature rise of more than 55°F above ambient shall be cause for rejection. Immediately after the temperature stabilizes, the digger shall be used to dig a series of holes not less than 12 feet deep. Each hole shall be dug in rapid succession, without moving the truck, and with spin-off time held to a minimum. For hard, frozen ground or other hard soil requiring picking for removal, not less than three holes shall be dug. For all other firm undisturbed soil conditions, not less than 5 holes shall be dug. When ledge rock or similar underground geological conditions prevent digging to a depth of

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12 feet, a greater number of holes dug to a total equivalent depth shall be acceptable. A temperature rise in the hydraulic fluid of more than 55°F above ambient in an ambient of 125°F, or equivalent temperature rise in other ambients interpolated to 125°F, at any time during the test, shall be cause for rejection.

* 4.3.8 Dielectric tests. Dielectric tests shall be conducted at line-to-ground voltages specified in 3.6.3.1 and as follows. Except for the basket immersion test, all tests shall be conducted with all components assembled for operational use.

(a) Vehicle under and parallel to and not more than 12 inches away from an energized conductor bar with ground cable attached to the vehicle, and to test unit. Apply required voltage when the conductor bar is not attached to the vehicle. The total internal leakage in milliamperes indicated on the test unit shall be observed.

(b) Secure a cable from the energized conductor bar to the following areas and apply the required voltage. The total leakage in milliamperes, if any, will be indicated on the test unit. The difference of (a) and (b) shall not exceed 1.0 milliampere.

(1) Basket bottom, through the insulated jib or insulated telescopic section to ground, with and without basket liner

(2) Basket front, through the insulated jib or insulated telescopic section to ground, with basket liner

(3) Basket attachment, through the jib or insulated telescopic section to ground, with and without basket liner.

(c) Disconnect the ground cable and attach to the basket to test as follows:
Secure a cable from the energized conductor bar to the basket front and test, then the bottom, with and without basket liner, and apply the required voltage. The total leakage in milliamperes indicated on the test unit shall be observed for each test.

(d) Basket immersion test, with basket immersed in a brine solution, and the interior of the basket filled to within a depth that is practical to prevent arcing.

4.3.9 Basket control tests. Using the basket controls only, the derrick boom shall be rotated, extended, retracted, elevated and lowered both fast and slow and in random sequence to assure responsive, trouble-free operation. While the boom is operated from the basket controls, the operator at the lower console shall be capable of taking

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over boom movement whenever desired. Take-over shall be smooth, precise and without appreciable oscillation. Upon release of basket control deadman switch, overtravel and oscillation shall be limited to a maximum of 4 inches of travel and 4 oscillations respectively at feather speed.

* 4.3.9.1 Fiber optic (on vehicle) system tests. The following tests shall be performed with the fiber optic system installed on the vehicle, as specified herein. Each test (except 4.5.9.1.4) shall be performed with the fiber optic and hydraulic systems operating. At the end of each test (see 4.3.9.1.1 through 4.3.9.4.7), the boom shall be operated from the basket controls not less than 2 times. Operation shall be smooth and nonerratic in all functions.

* 4.3.9.1.1 External moisture test. The transmitter, the receiver and the hydraulic valve actuating solenoids shall be sprayed thoroughly for not less than five minutes from the 1/2-inch hose at a pressure of not less than 30 pounds per square inch (psi) from a nozzle distance of approximately 3 feet. At the end of the test, check inside the enclosures of the transmitter and receiver for any water. Accumulation shall be negligible (equivalent to not more than 10 shots from a spray bottle).

* 4.3.9.1.2 Internal moisture test. Remove the transmitter and receiver covers. Spray not less than 10 shots of water onto the electronic components of each enclosure using a hand held spray bottle.

* 4.3.9.1.3 Heat test. The following test shall be performed immediately after moisture tests (see 4.3.9.1.1 and 4.3.9.1.2) without the removal of any moisture present. With the transmitter and receiver enclosure covers reinstalled, means shall be provided to raise the interior temperatures of the transmitter and receiver to not less than plus 145°F. A suitable temperature measuring device shall be used to insure that the interior temperature is held for not less than 10 minutes.

* 4.3.9.1.4 Dielectric test. A dielectric test shall be performed in accordance with 4.3.8(c).

* 4.3.9.1.5 Electric dump test. With the vehicle engine running at idle, disconnect and reconnect the vehicle battery not less than five times. The fiber optics system shall continue to function properly.

* 4.3.9.1.6 Low voltage test. With the vehicle engine running at idle, turn on lights, heaters, fans, and all other vehicle electrical components to achieve a full electrical load condition. The fiber optic system shall function properly.

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* 4.3.9.1.7 Receiver power interrupt test. With the "manual/bucket" switch on the receiver in the "manual" position and the vehicle engine running at idle, disconnect power from the receiver for not less than 2 seconds. Reconnect power to the receiver. Operate the fiber optic system from the receiver. Operation shall return to normal when power is restored.

* 4.3.9.2 Fiber optic (off-vehicle) system tests. The following tests shall be performed by the fiber optic system manufacturer. Test shall be performed on each system, and on each transmitter and receiver. Tests shall be performed after a burn-in period of not less than 48 hours with loads simulating operation on-vehicle.

* 4.3.9.2.1 Life cycle test. With simulated loads connected, operate the system for not less than 500 cycles. The system shall operate properly in all functions.

* 4.3.9.2.2 Temperature tests. The following tests shall be performed at ambient temperature, at plus 145°F, at minus 40°F, and again at ambient temperature. The test results in the plus 145°F tests, minus 40°F tests, and in the final ambient temperature test shall not vary from the results of the initial ambient temperature tests. After the initial ambient temperature tests, raise the transmitter and receiver temperature to plus 145°F in not less than 30 minutes. After the plus 145°F tests, lower the temperature to minus 40°F in not less than 30 minutes. After the minus 40°F tests, raise the temperature to ambient in not more than 20 minutes.

(a) Interaction test. Operate system. There shall be no interaction from one input or output with any other input or output or any combination thereof
(b) Proportional current test. Operate system. Valve output current shall not vary as a function of receiver input voltage range or valve resistance variation
(c) On-off test. Turn system on, operate, turn system off and then on again. System shall function properly when turned back on
(d) Dynamic operation test. Operate system. All analog and digital, input and output functions shall operate in accordance with requirements specified (see 3.6.7.4)
(e) Linearity test. Operate system. All analog channel outputs shall be proportional to their respective command inputs.

4.3.10 Overhead guard test. The guard shall be tested or calculations shall be performed to show conformance to 3.6.7.2.

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4.3.11 Section modulus. When a rigid mounting system for the cabinet-floor assembly is used, the chassis frame rails and the subframe shall be measured and the section modulus computed by the contractor to determine conformance to 3.5.2.2.

4.3.12 Production sample. The first vehicle shall remain at the manufacturing facility as a production sample and shall be the last vehicle shipped on the contract. The contractor shall maintain the vehicle in a serviceable condition for the duration of the contract.

* 4.3.12.1 Refurbishment of first vehicle. Prior to delivery, the approved first vehicle shall be refurbished to new condition. Refurbishment shall include, as a minimum, any repairs or replacement necessitated by wear and tear from testing; replacement of all normal service items; and repainting as required to give new condition appearance.

* 4.3.13 Failure. Failure of the first vehicle to meet requirements of the contract shall be cause for the Government to refuse acceptance of all vehicles under contract until corrective action has been taken. Following correction, testing may resume. Any tests performed prior to failure shall be repeated as deemed necessary by the Government.

4.4 Inspection of production vehicles. The contractor's inspection system shall as a minimum assure that the vehicle conforms to the physical and dimensional requirements and is capable of meeting performance requirements contained herein. Acceptance tests shall include an examination of the product and performance checks as specified in 4.3.5(a) through (e). For each vehicle under contract, the contractor shall make available to the Government, at the point of final acceptance, records acceptable to the Government indicating that these acceptance tests and the servicing and adjusting required by 3.10 have been accomplished.

5. PACKAGING

5.1 Vehicle processing. The vehicle shall be processed for shipment, from the manufacturer's plant to the initial receiving activity, in accordance with the manufacturer's standard commercial practice.

6. NOTES

6.1 Intended use. The vehicles covered by this specification are for general nontactical use by the Government in the performance of the maintenance and construction tasks indicated.

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6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) A first article and a first article test, if required (see 3.1).
(c) Identification of appropriate service for painting and marking (see 3.1.1.1).
(d) Concealed markings, if required (see 3.1.1.1).
(e) Rustproofing, if required (see 3.1.1.2).
(f) Tropical rustproofing, if required (see 3.1.1.2).
(g) Radiator shutters, if required (see 3.1.1.13).
(h) Keys common to all trucks, if required (see 3.1.1.14).
(i) Silicone rubber radiator and heater hoses, if required (see 3.4.1.7).
(j) Power plant heaters and a fuel warmer, if required (see 3.4.1.8).
(k) If a battery heater is not required (see 3.4.1.8).
(l) Traction control, if required (see 3.4.9.2).
(m) Disc type wheels, if required (see 3.4.10).
(n) Bias ply tires, if required (see 3.4.10.1).
(o) Spare wheel or rim and carrier, if required (see 3.4.10.3).
(p) Spare wheel or rim, stowed on vehicle, if required (see 3.4.10.3).
(q) Spare tire assembly, if required (see 3.4.10.3.1).
(r) Forward tilting hood and fender assembly, if required (see 3.4.12).
(s) Individual driver and passenger seats, if required (see 3.4.12).
(t) Tools, if required (see 3.4.16.1).
(u) Air operated horn, in addition to the electric horn, if required (see 3.4.21).
(v) Winch, rear mounted, if required (see 3.5.6).
(w) Two one-man baskets, if required (see 3.6.3.3).
(x) If augers are not required (see 3.6.6.1).
(y) Hydraulic fluid conforming to MIL-H-83282, if required (see 3.6.8.4.1).
(z) Basket outlet and hose reel, if required (see 3.6.10.1).
(aa) Basket hydraulic chain saw, if required (see 3.6.10.2).
(ab) Basket hydraulic chain saw, pole mounted, if required (see 3.6.10.3).
(ac) Basket hydraulic circular saw, pole mounted, if required (see 3.6.10.4).
(ad) Basket hydraulic impact wrench and drill, if required (see 3.6.10.5).
(ae) Basket hydraulic pruner, pole mounted, if required (see 3.6.10.6).

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* 6.3 Performance prediction. SAE Truck Ability Prediction Procedure computations and computations for low speed and maximum geared speed will be required by the contract. The SAE Work Sheet item 1 should include vehicle model number and engine model number. Unless other conditions are cited in the contract, computations should be made for normal atmospheric pressure, normal ambient air temperature, and still dry air. The factors to be used in predicting truck ability (see 3.3.1.1) are established as follows for the corresponding SAE Truck Ability Prediction Procedure Tables:

Table 1 - Tire Factor. This factor relates to the size of tires furnished by the contractor in accordance with this specification.
Table 2 - Altitude Factor. 1.00
Table 3 - Rolling Factor. 1.613
Table 4 - Area Factor. 0.173
Table 5 - Velocity Factor. 250.0
Table 6 - Altitude Factor. 1.00
Table 7 - Chassis Friction Horsepower. Use applicable power unit GVW (to nearest, higher, 1,000 pounds) and the engine rpm (to nearest 100 revolutions) which is required for 50 mph geared speed. For GVW and engine speed beyond the range of this table, factors shall be extrapolated.
Table 8 - Grade Factor. 0.75
Table 8A - Correction Factor. Not required.
Table 9 - Road Factor. 0.0.

* 6.4 Subject term (key word) listing.
Aerial basket
Air transportability
Derrick, rear of cab
Fiber optic control system
Pin-on jib
Telescopic hydraulic derrick
Truck, commercial
Truck, maintenance
Truck, nontactical.

6.5 Identification of changes. The margins of this specification are marked with asterisks (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no

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liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content regardless of the marginal notations and relationship to the last previous issue.

Custodians:
Army - AT
Air Force - 99

Review activity:
Air Force - 84

Preparing activity:
Army - AT

Project No. 2320-0326

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE (MIL-T-62089E) TRUCK, MAINTENANCE: With Rotating Hydraulic Derrick, Air Transportable, DED, 39,500 GVW, 6x4, Commercial

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

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NOTICE OF INACTIVATION
FOR NEW DESIGN

INCH-POUND

MIL-T-62089E
NOTICE 1
15 December 1995

MILITARY SPECIFICATION

TRUCK MAINTENANCE: WITH ROTATING HYDRAULIC DERRICK, AIR TRANSPORTABLE, DIESEL ENGINE DRIVEN, 39,500 POUNDS GVW, 6X4, COMMERCIAL

This notice should be filed in front of MIL-T-62089E, dated 27 May 1988.

MIL-T-62089E is inactive for new design and is no longer used, except for replacement purposes.

Preparing activity:
Army - AT

(Project 2320-0644)

AMSC N/A FSC 2320

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